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Tang et al.

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(54) **IMAGE CAPTURING LENS ASSEMBLY,
IMAGE CAPTURING DEVICE AND MOBILE
TERMINAL**

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G02B 13/00 (2006.01)
G02B 3/04 (2006.01)
G02B 13/18 (2006.01)

(52) **U.S. Cl.**
CPC **G02B 13/0045** (2013.01); **G02B 3/04**
(2013.01); **G02B 13/18** (2013.01)

(58) **Field of Classification Search**
CPC G02B 13/0045; G02B 13/18
See application file for complete search history.

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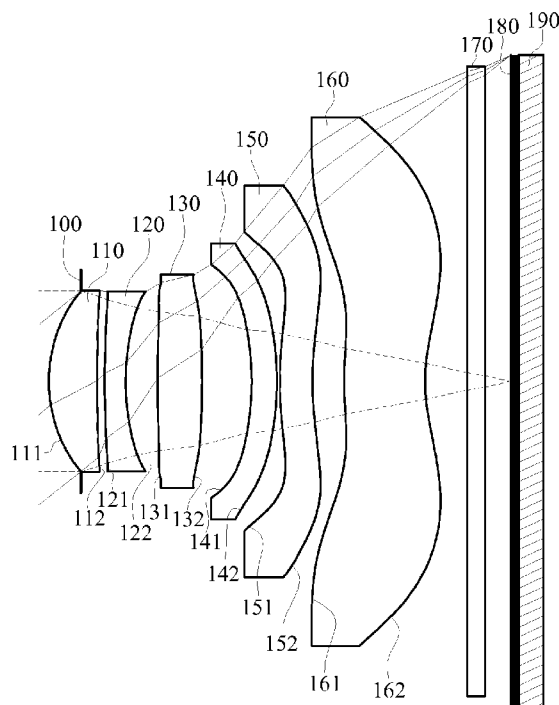
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(57) **ABSTRACT**

An image capturing lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, and a sixth lens element. The first lens element with positive refractive power has an object-side surface being convex in a paraxial region thereof. The second and third lens elements have refractive power. The fourth lens element with refractive power has an object-side surface being concave in a paraxial region thereof. The fifth lens element with refractive power has an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof. The sixth lens element with refractive power has an image-side surface being concave in a paraxial region thereof. The image capturing lens assembly has six lens elements with refractive power.

25 Claims, 18 Drawing Sheets



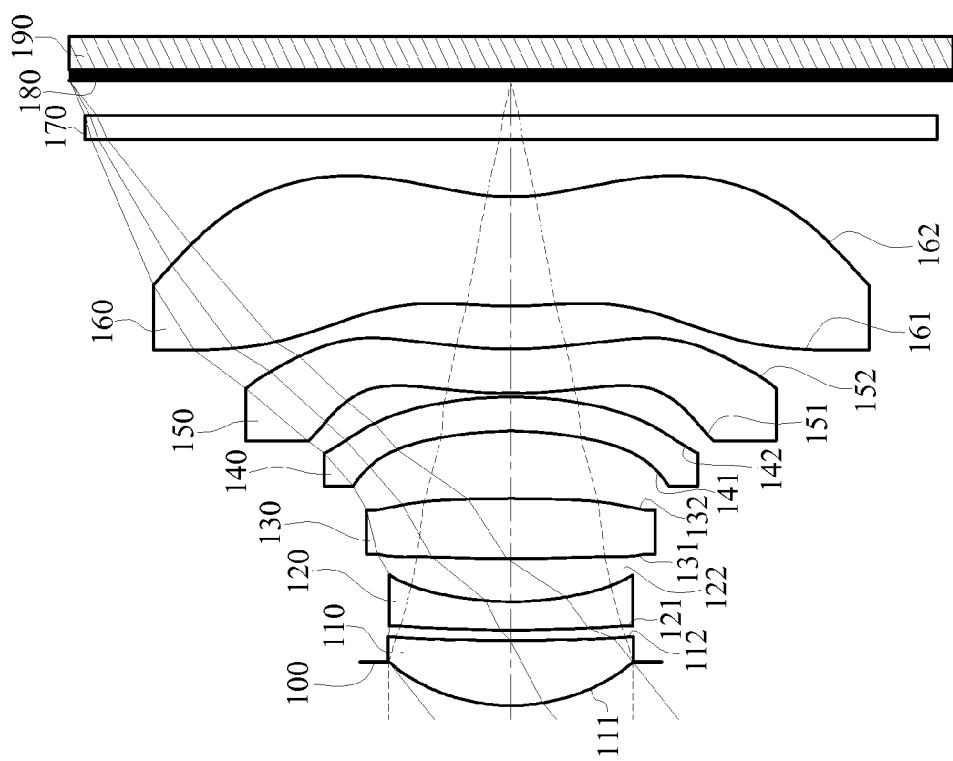


FIG. 1

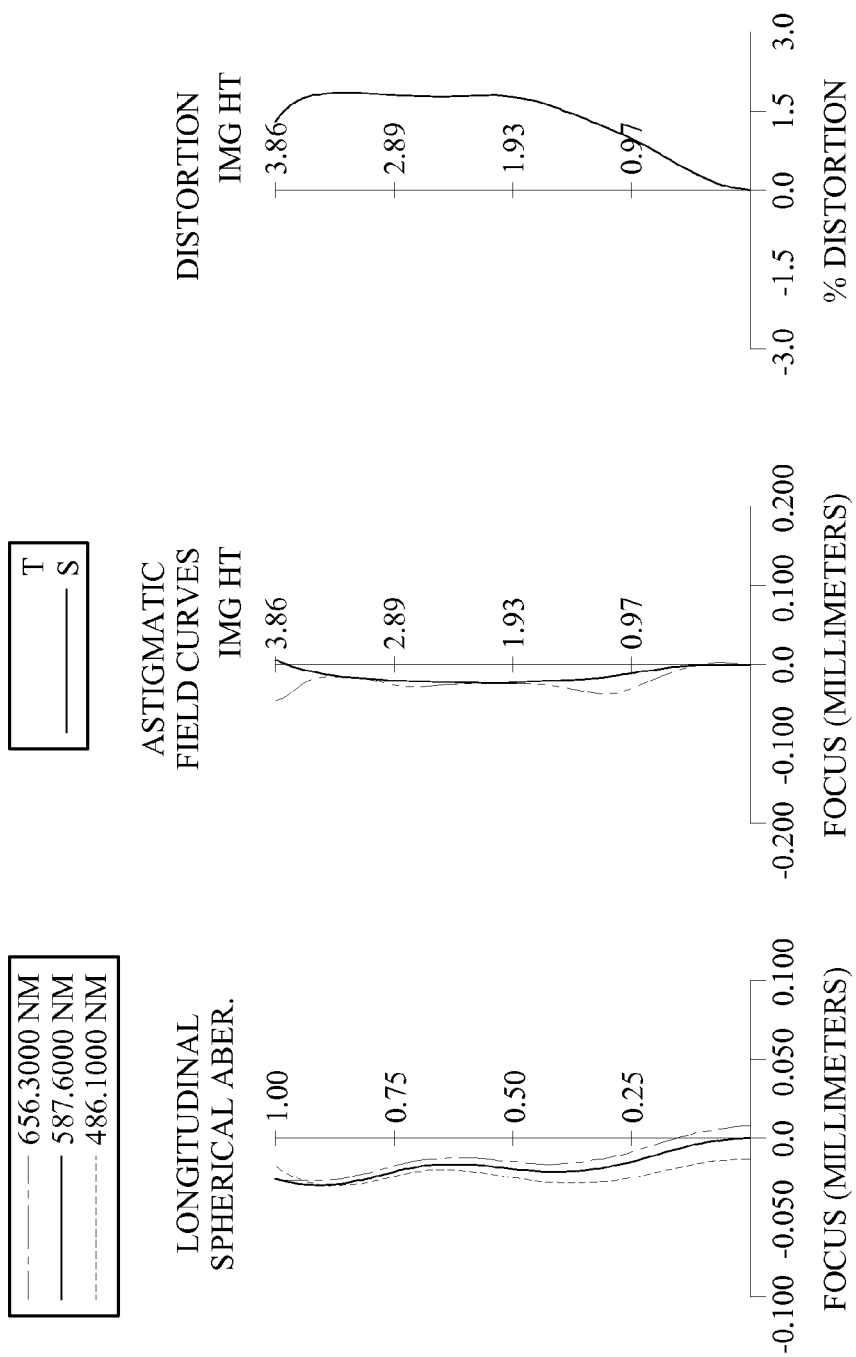


FIG.2

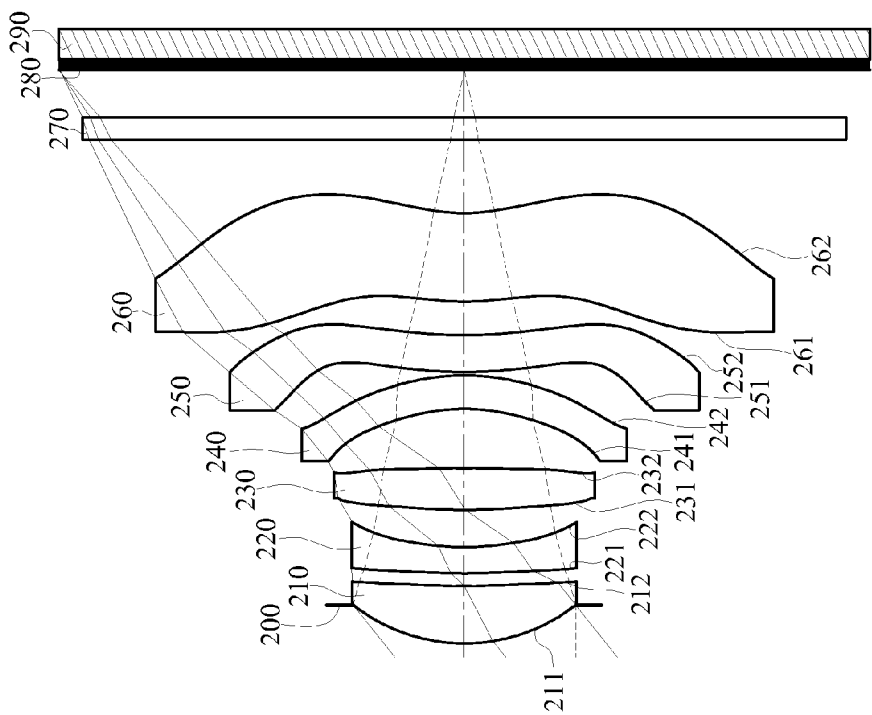


FIG.3

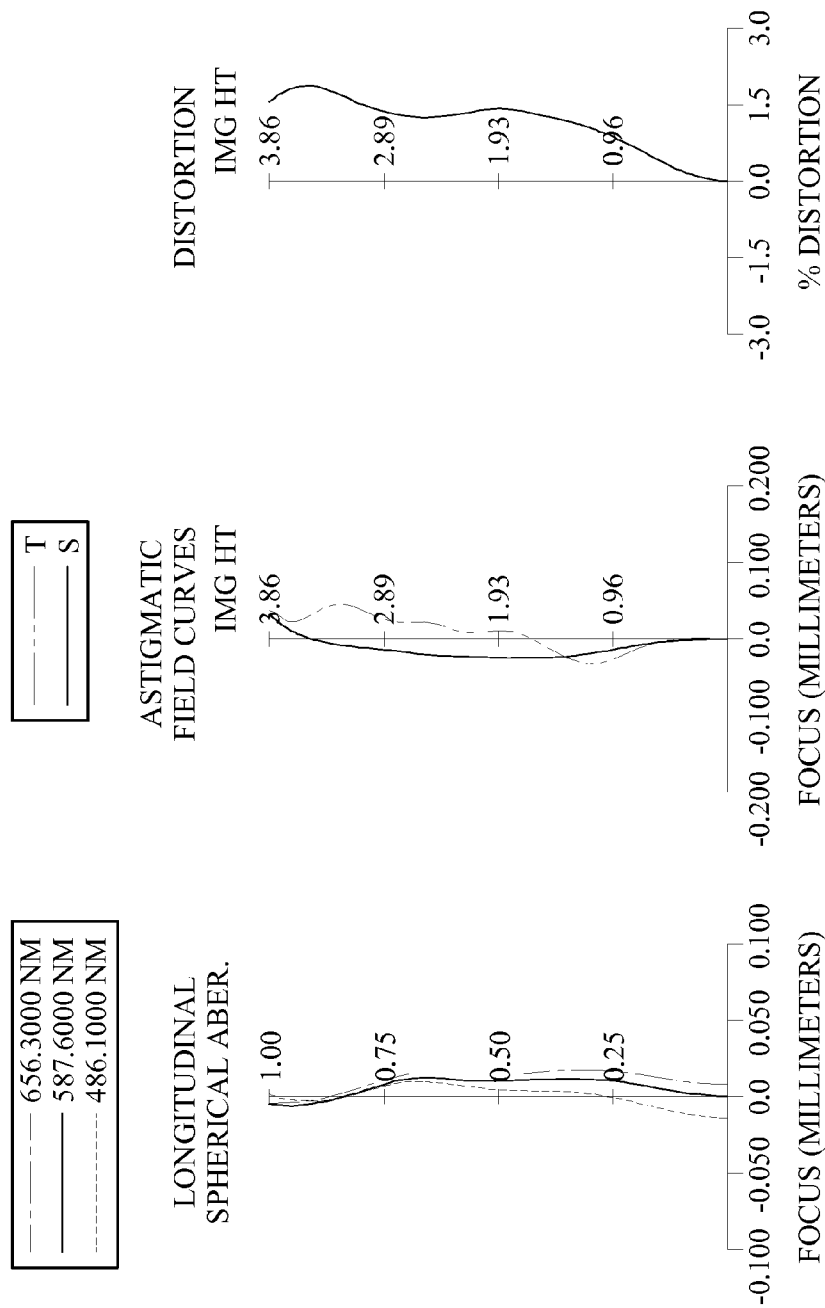


FIG. 4

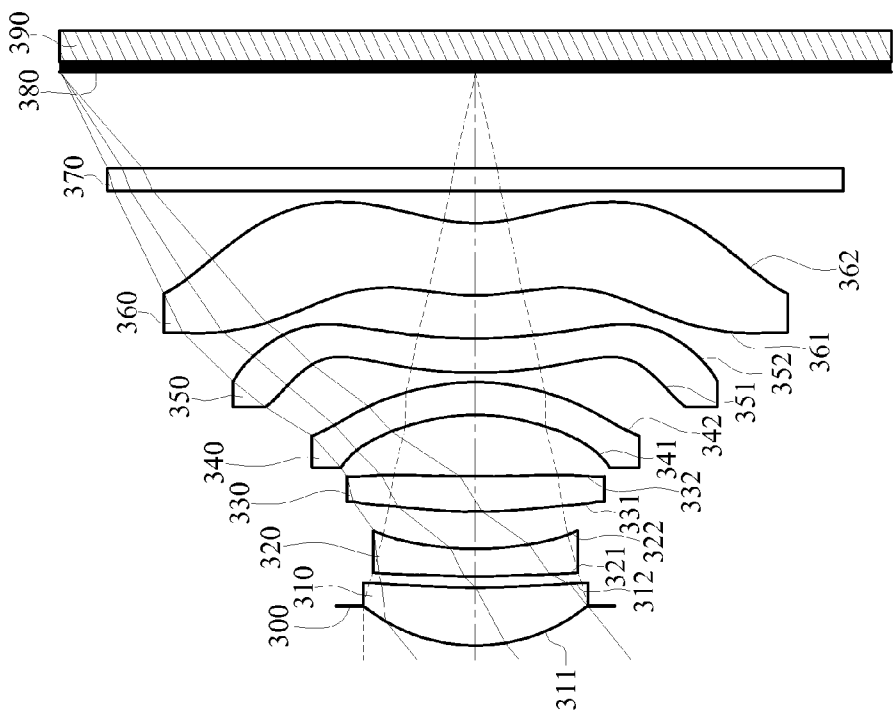


FIG. 5

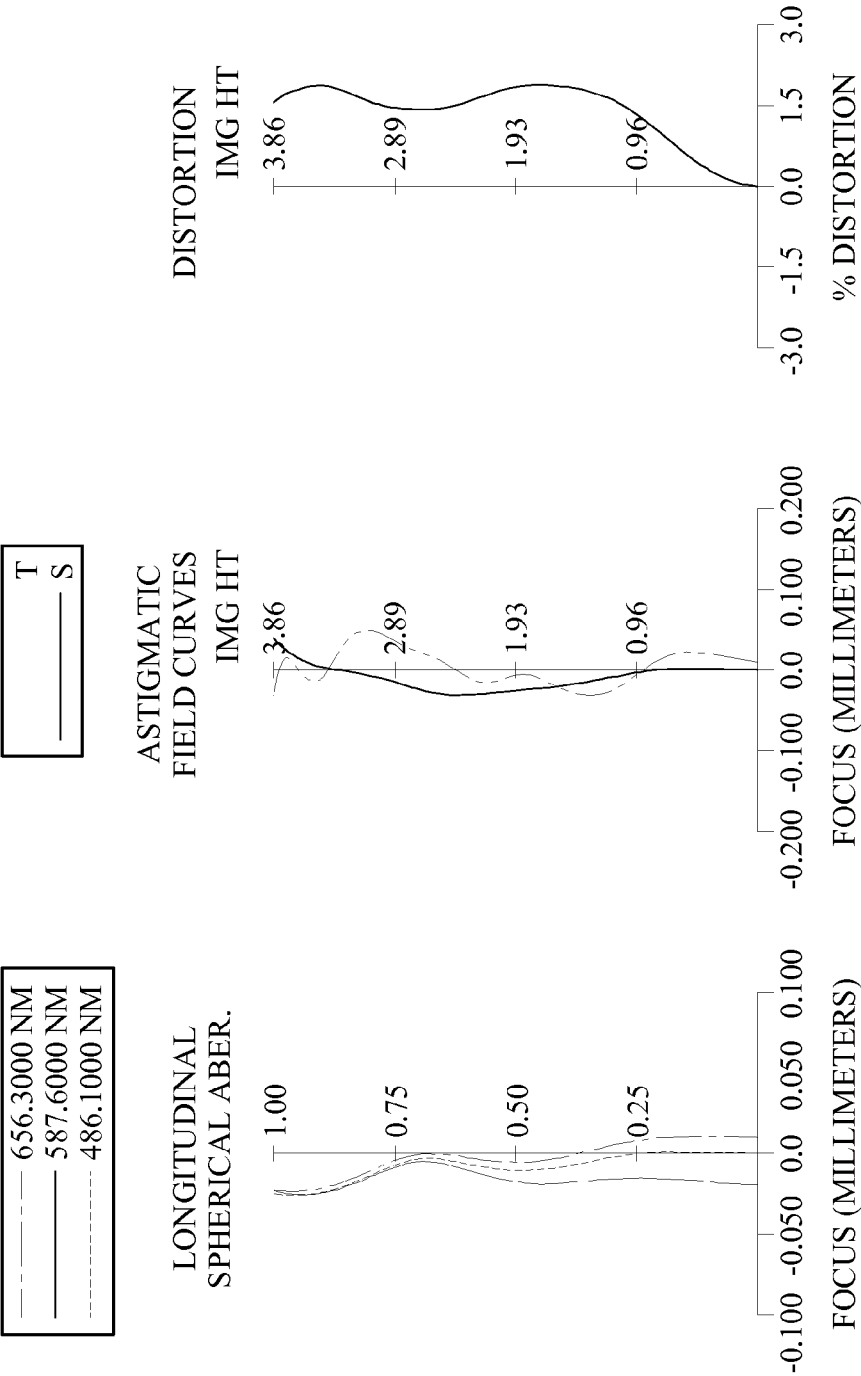


FIG.6

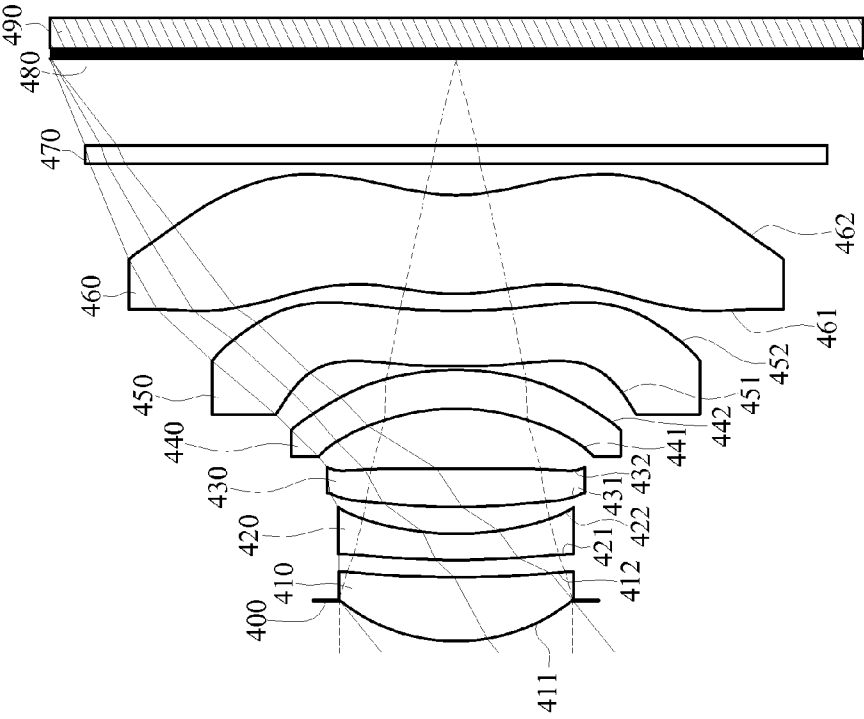
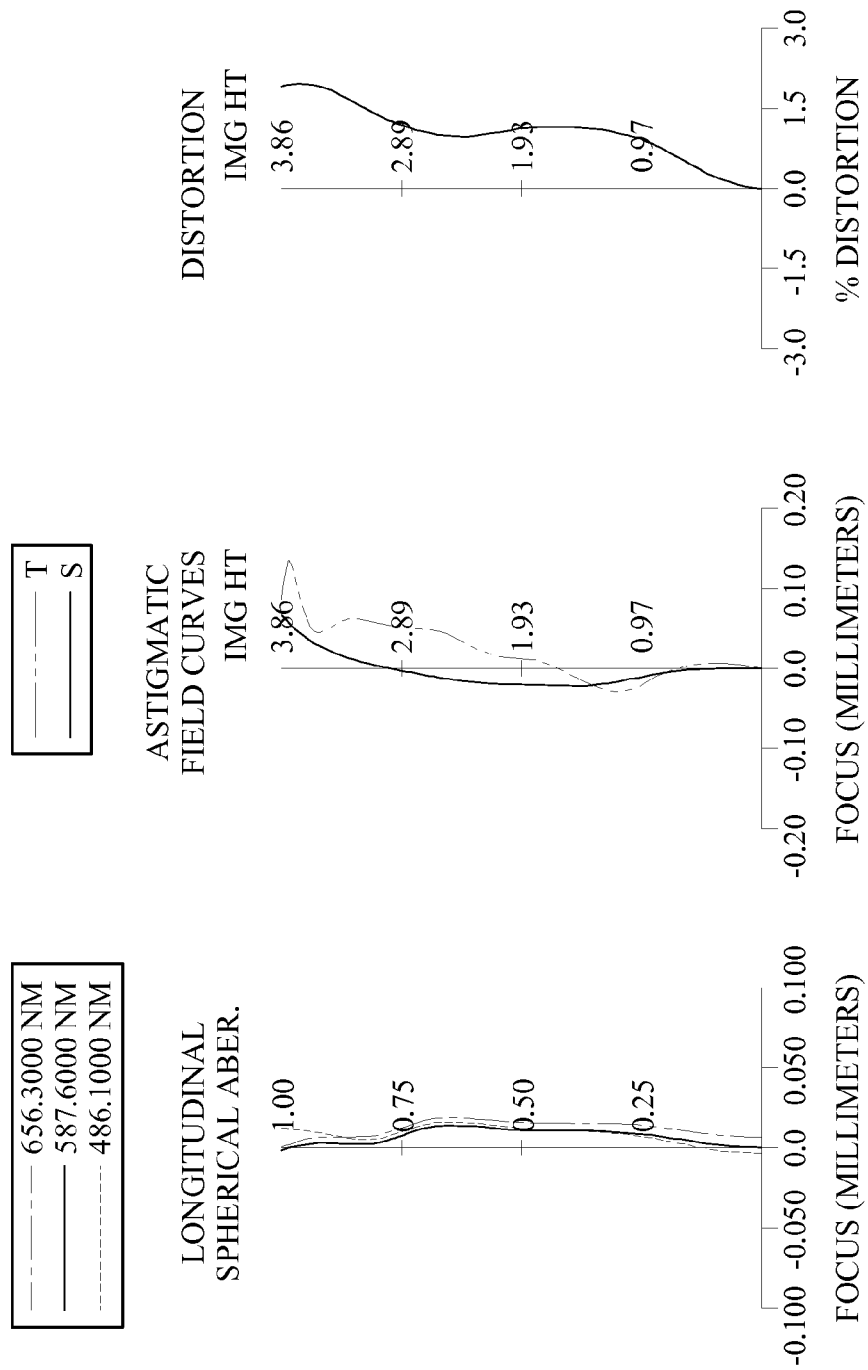


FIG. 7



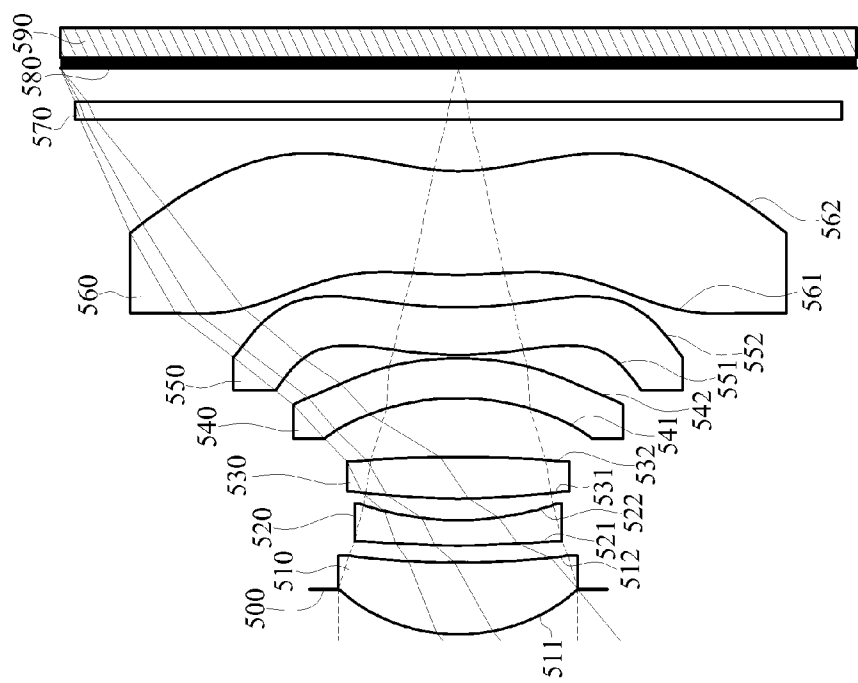


FIG. 9

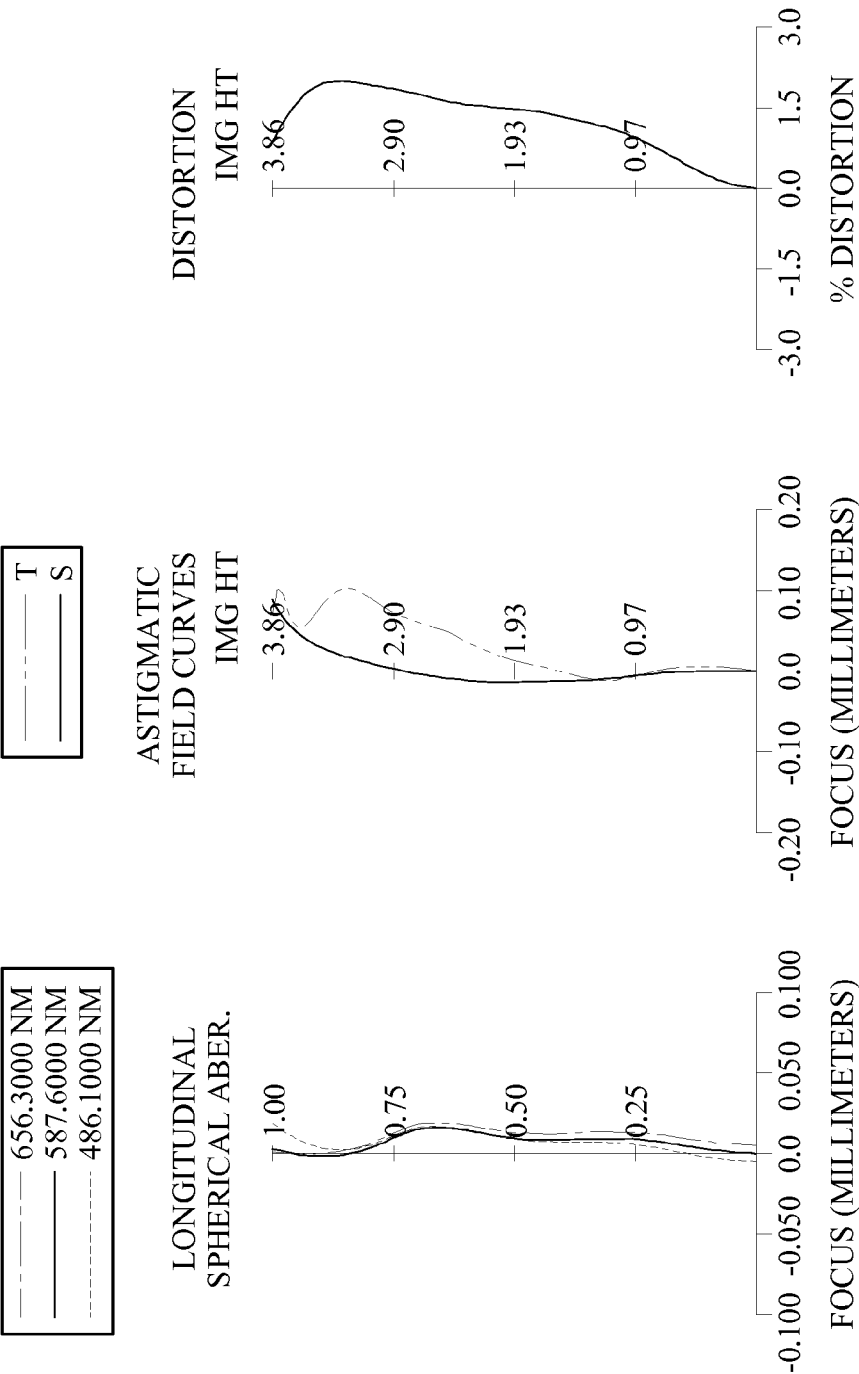


FIG.10

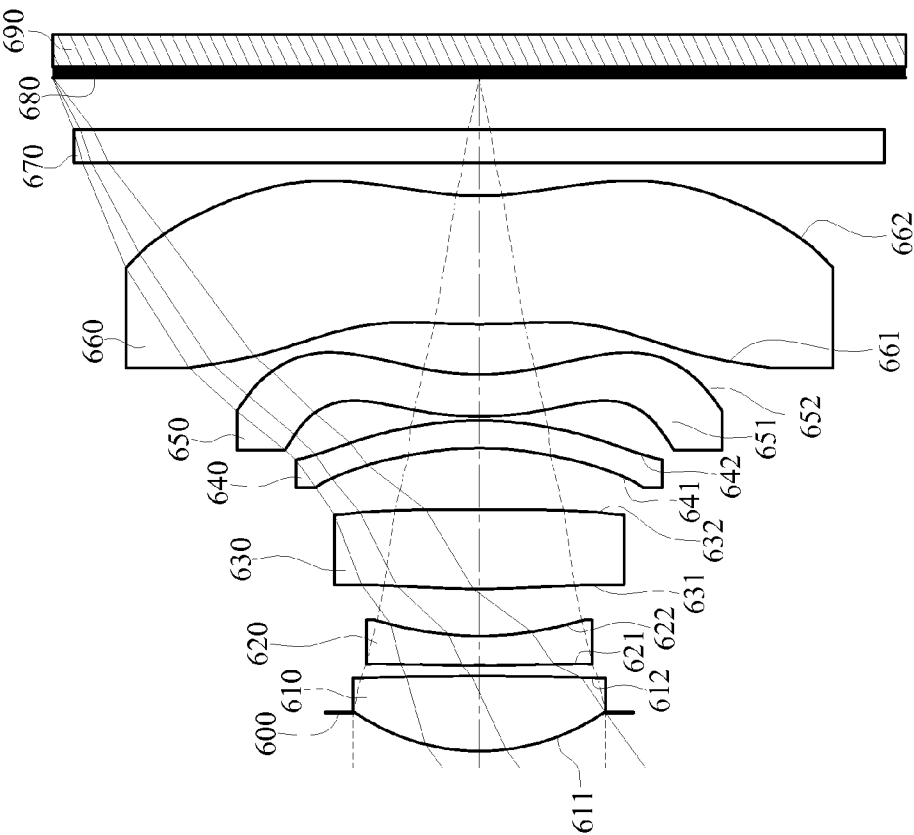


FIG.11

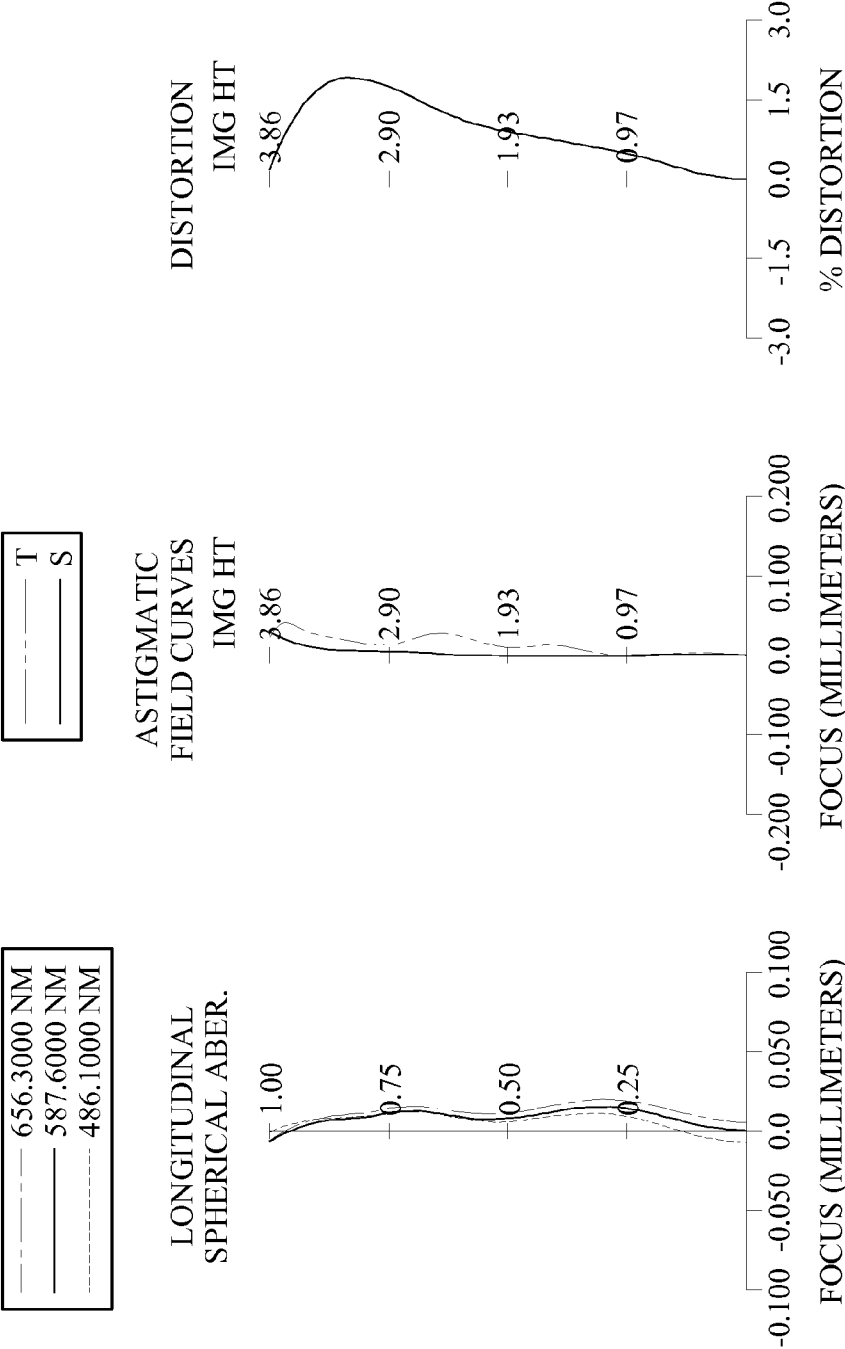


FIG.12

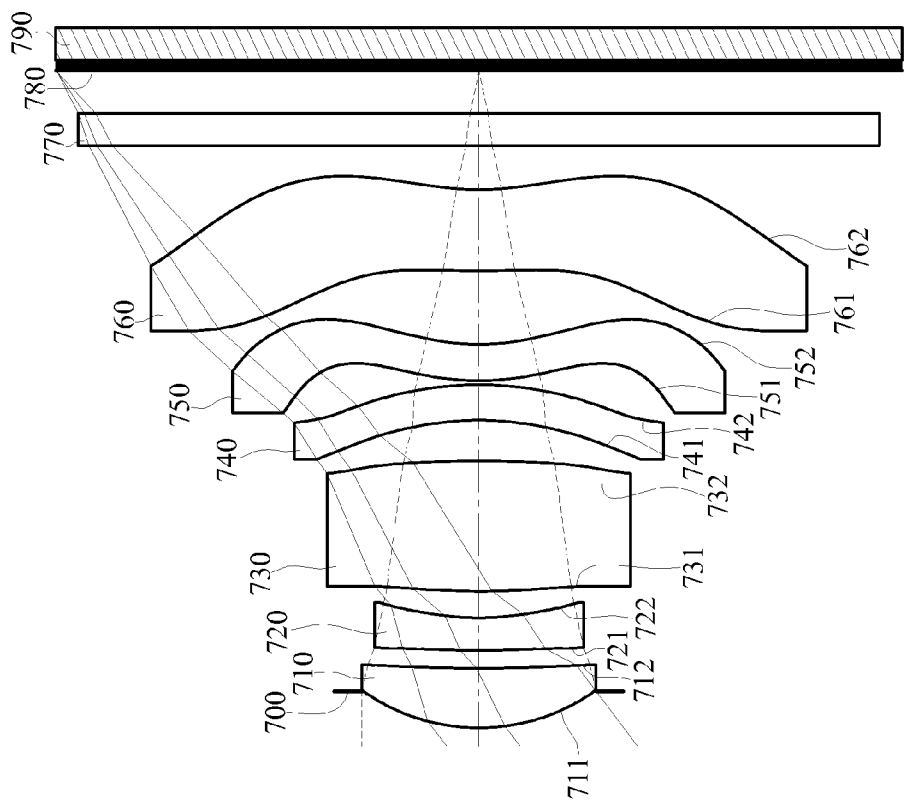


FIG.13

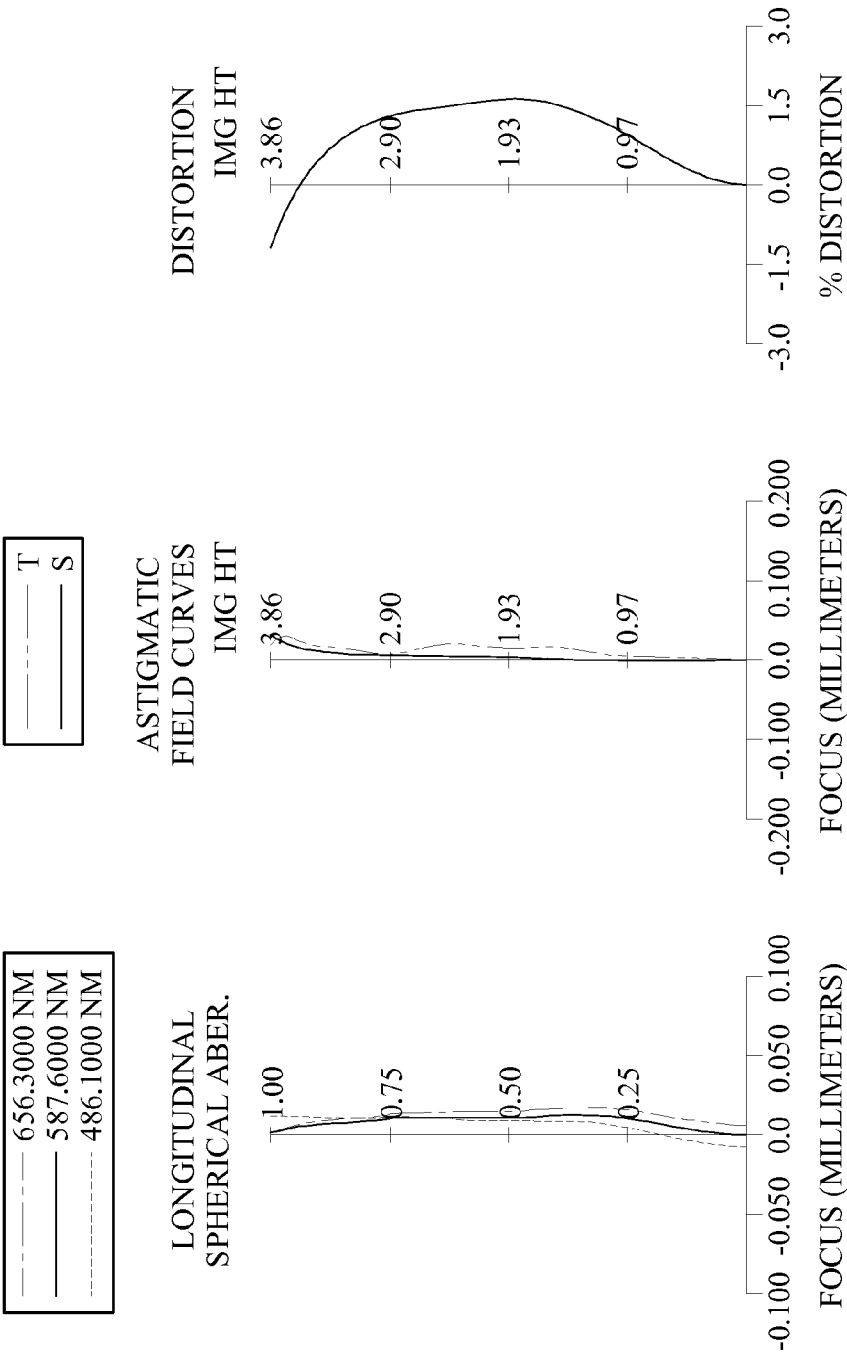


FIG.14

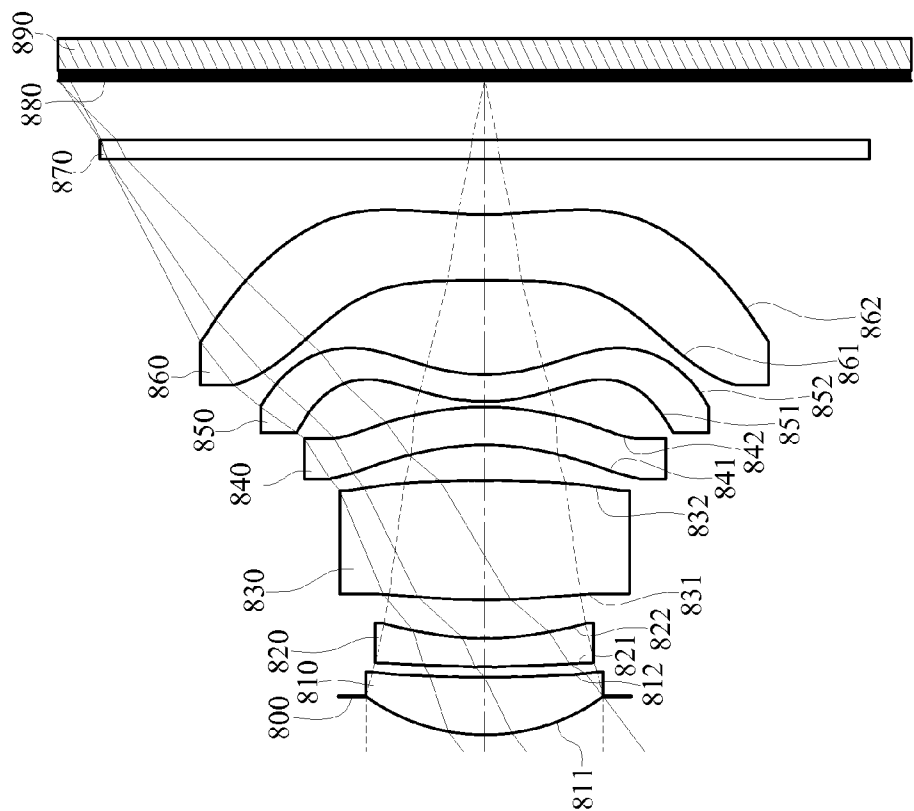


FIG.15

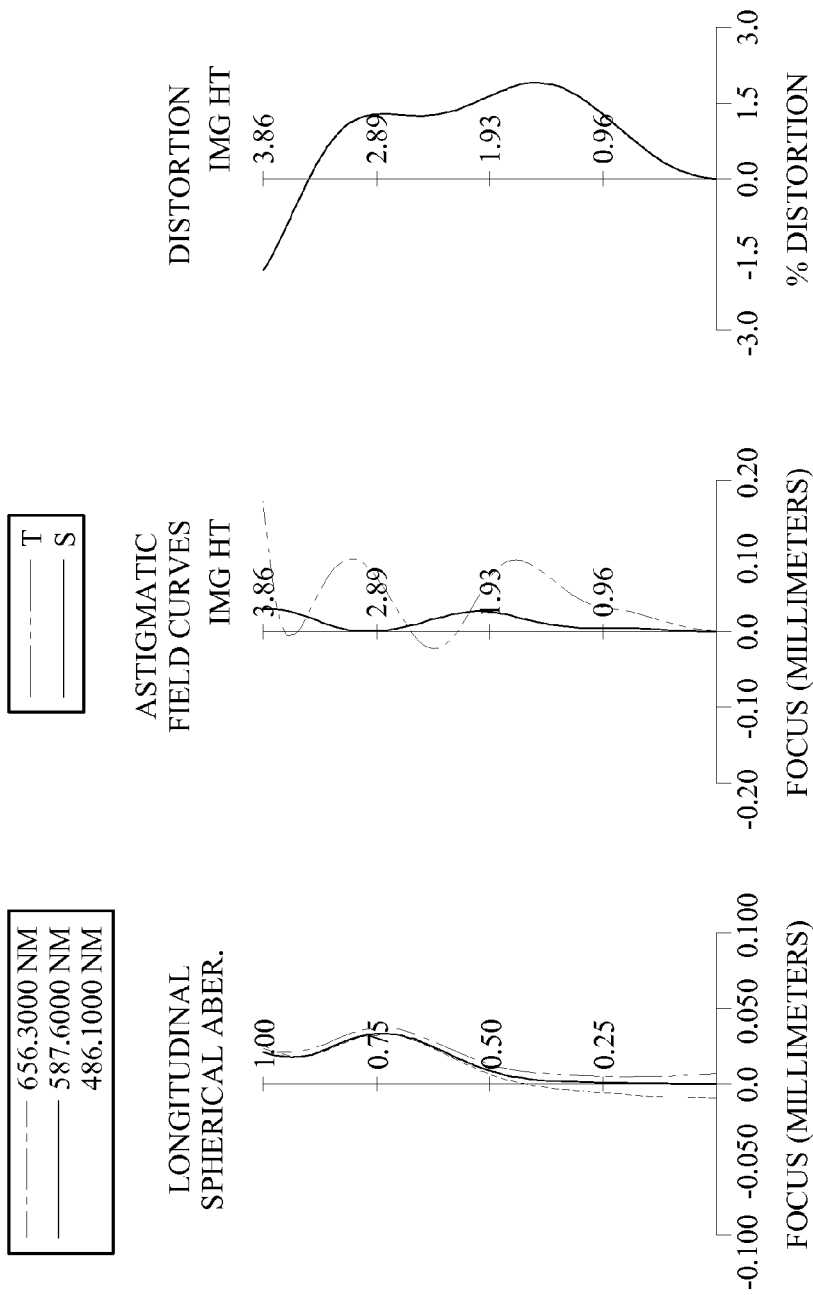


FIG.16

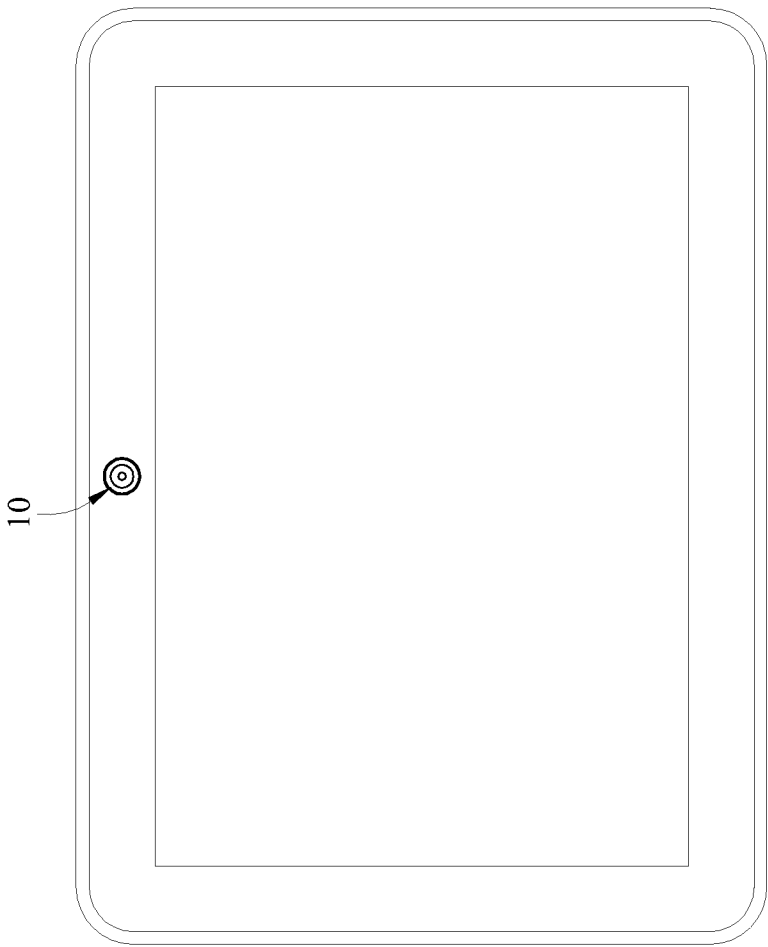


FIG.18

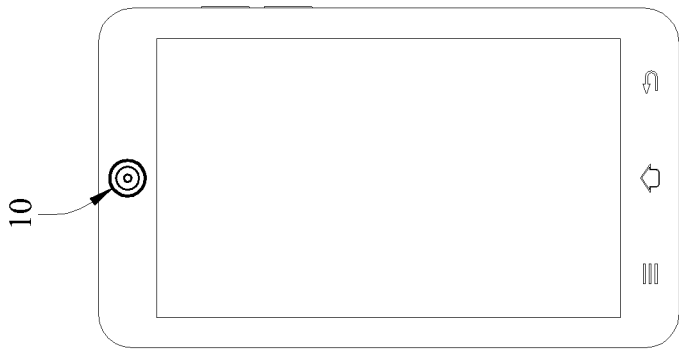


FIG.17

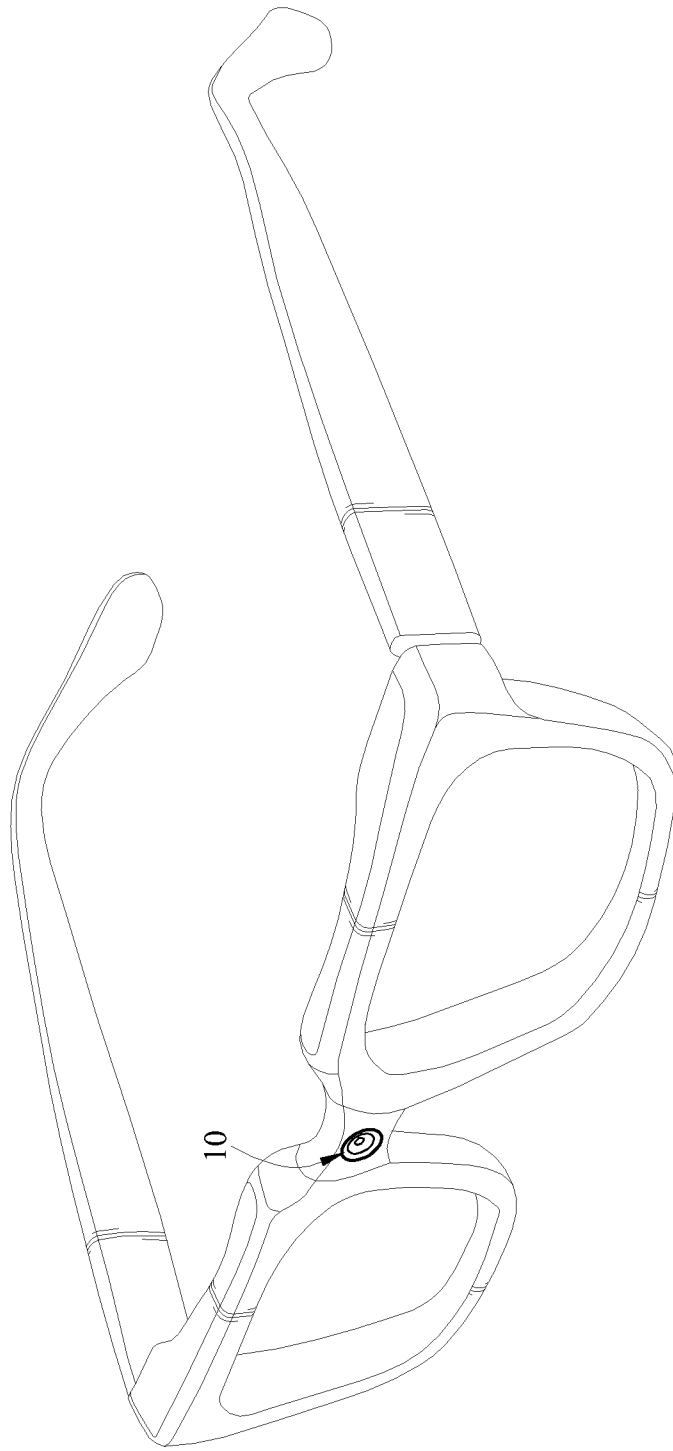


FIG. 19

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IMAGE CAPTURING LENS ASSEMBLY, IMAGE CAPTURING DEVICE AND MOBILE TERMINAL

RELATED APPLICATIONS

This application claims priority to Taiwan Application Serial Number 102143877, filed Nov. 29, 2013, which is incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

The present disclosure relates to an image capturing lens assembly, an image capturing device and a mobile terminal. More particularly, the present disclosure relates to a compact image capturing lens assembly applicable to a mobile terminal.

2. Description of Related Art

In recent years, with the popularity of mobile terminals having camera functionalities, the demand of miniaturized optical systems has been increasing. The sensor of a conventional optical system is typically a CCD (Charge-Coupled Device) or a CMOS (Complementary Metal-Oxide-Semiconductor) sensor. As the advanced semiconductor manufacturing technologies have allowed the pixel size of sensors to be reduced and compact optical systems have gradually evolved toward the field of higher megapixels, there is an increasing demand for compact optical systems featuring better image quality.

A conventional optical system employed in a portable electronic product mainly adopts a five-element lens structure. Due to the popularity of mobile terminals with high-end specifications, such as smart phones, tablet personal computers and wearable apparatus, the requirements for high resolution and image quality of present compact optical systems increase significantly. However, the conventional optical systems cannot satisfy these requirements of the compact optical systems.

Other conventional compact optical systems with six-element lens structure enhance image quality and resolution. However, the surface shape and refractive power of the lens elements are not favorable for reducing the sensitivity of the optical systems which might thereby result in worse image quality.

SUMMARY

According to one aspect of the present disclosure, an image capturing lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, and a sixth lens element. The first lens element with positive refractive power has an object-side surface being convex in a paraxial region thereof. The second lens element has refractive power. The third lens element has refractive power. The fourth lens element with refractive power has an object-side surface being concave in a paraxial region thereof. The fifth lens element with refractive power has an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof, wherein the image-side surface of the fifth lens element has at least one convex shape in an off-axis region thereof, and the object-side surface and the image-side surface of the fifth lens element are aspheric. The sixth lens element with refractive power has an image-side surface being concave in a paraxial region thereof, wherein the image-side surface of the sixth

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lens element has at least one convex shape in an off-axis region thereof, and an object-side surface and the image-side surface of the sixth lens element are aspheric. The image capturing lens assembly has a total of six lens elements with refractive power. When a focal length of the third lens element is $f3$, a focal length of the fourth lens element is $f4$, an axial distance between the third lens element and the fourth lens element is $T34$, a central thickness of the fourth lens element is $CT4$, a central thickness of the fifth lens element is $CT5$, and a central thickness of the sixth lens element is $CT6$, the following conditions are satisfied:

$$|f3/f4| < 0.50;$$

$$0.9 < T34/CT4; \text{ and}$$

$$1.6 < CT6/CT5.$$

According to another aspect of the present disclosure, an image capturing device includes the image capturing lens assembly according to the aforementioned aspect and an image sensor.

According to still another aspect of the present disclosure, a mobile terminal includes the image capturing device according to the aforementioned aspect.

According to yet another aspect of the present disclosure, an image capturing lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, and a sixth lens element. The first lens element with positive refractive power has an object-side surface being convex in a paraxial region thereof. The second lens element has refractive power. The third lens element has refractive power. The fourth lens element with refractive power has an object-side surface being concave in a paraxial region thereof. The fifth lens element with refractive power has an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof, wherein the image-side surface of the fifth lens element has at least one convex shape in an off-axis region thereof, and the object-side surface and the image-side surface of the fifth lens element are aspheric. The sixth lens element with negative refractive power has an image-side surface being concave in a paraxial region thereof, wherein the image-side surface of the sixth lens element has at least one convex shape in an off-axis region thereof, and an object-side surface and the image-side surface of the sixth lens element are aspheric. The image capturing lens assembly has a total of six lens elements with refractive power. When a focal length of the third lens element is $f3$, a focal length of the fourth lens element is $f4$, a focal length of the fifth lens element is $f5$, a focal length of the sixth lens element is $f6$, an axial distance between the third lens element and the fourth lens element is $T34$, and a central thickness of the fourth lens element is $CT4$, the following conditions are satisfied:

$$|f3/f4| < 0.50;$$

$$0.9 < T34/CT4; \text{ and}$$

$$-0.65 < f6/f5 < 0.$$

According to still yet another aspect of the present disclosure, an image capturing device includes the image capturing lens assembly according to the foregoing aspect and an image sensor.

According to a further aspect of the present disclosure, a mobile terminal includes the image capturing device according to the foregoing aspect.

BRIEF DESCRIPTION OF THE DRAWINGS

The disclosure can be more fully understood by reading the following detailed description of the embodiments, with reference made to the accompanying drawings as follows:

FIG. 1 is a schematic view of an image capturing device according to the 1st embodiment of the present disclosure;

FIG. 2 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 1st embodiment;

FIG. 3 is a schematic view of an image capturing device according to the 2nd embodiment of the present disclosure;

FIG. 4 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 2nd embodiment;

FIG. 5 is a schematic view of an image capturing device according to the 3rd embodiment of the present disclosure;

FIG. 6 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 3rd embodiment;

FIG. 7 is a schematic view of an image capturing device according to the 4th embodiment of the present disclosure;

FIG. 8 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 4th embodiment;

FIG. 9 is a schematic view of an image capturing device according to the 5th embodiment of the present disclosure;

FIG. 10 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 5th embodiment;

FIG. 11 is a schematic view of an image capturing device according to the 6th embodiment of the present disclosure;

FIG. 12 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 6th embodiment;

FIG. 13 is a schematic view of an image capturing device according to the 7th embodiment of the present disclosure;

FIG. 14 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 7th embodiment;

FIG. 15 is a schematic view of an image capturing device according to the 8th embodiment of the present disclosure;

FIG. 16 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 8th embodiment;

FIG. 17 shows a mobile terminal according to the 1st embodiment;

FIG. 18 shows a mobile terminal according to the 2nd embodiment; and

FIG. 19 shows a mobile terminal according to the 3rd embodiment.

DETAILED DESCRIPTION

An image capturing lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, and a sixth lens element. The image capturing lens assembly has a total of six lens elements with refractive power.

The first lens element has positive refractive power, so that it provides the image capturing lens assembly with the positive refractive power as it needs to be. The first lens element has an object-side surface being convex in a paraxial region thereof, so that it is favorable for further enhancing the arrangement of the positive refractive power so as to reduce the total track length of the image capturing lens assembly.

The second lens element can have negative refractive power. Therefore, it is favorable for correcting the aberration of the first lens element with positive refractive power. Moreover, the second lens element can have an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof. Therefore, it is favorable for correcting the astigmatism of the image capturing lens assembly.

The third lens element can have positive refractive power. Therefore, it is favorable for balancing the arrangement of the positive refractive powers so as to reduce the sensitivity of the image capturing lens assembly.

The fourth lens element with refractive power has an object-side surface being concave in a paraxial region thereof. Therefore, it is favorable for correcting astigmatism.

The fifth lens element can have negative refractive power and has an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof, wherein the image-side surface of the fifth lens element has at least one convex shape in an off-axis region thereof. Therefore, it is favorable for further correcting the astigmatism and reducing the incident angle of the light projecting onto an image sensor so as to improve the responding efficiency of the image sensor.

The sixth lens element can have negative refractive power, an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof, wherein the image-side surface of the sixth lens element has at least one convex shape in an off-axis region thereof. Therefore, it is favorable for the principal point being positioned away from the image plane so as to reduce the total track length and effectively correct the aberration of the off-axis.

When a focal length of the third lens element is f_3 , and a focal length of the fourth lens element is f_4 , the following condition is satisfied: $|f_3/f_4| < 0.50$. Therefore, it is favorable for reducing the sensitivity of the image capturing lens assembly so as to obtain better image quality.

When an axial distance between the third lens element and the fourth lens element is T_{34} , and a central thickness of the fourth lens element is CT_4 , the following condition is satisfied: $0.9 < T_{34}/CT_4$. Therefore, it is favorable for assembling the image capturing lens assembly and maintaining a proper total track length thereof. In some embodiments, the following condition is satisfied: $1.25 < T_{34}/CT_4 < 2.75$.

When a central thickness of the fifth lens element is CT_5 , and a central thickness of the sixth lens element is CT_6 , the following condition is satisfied: $1.6 < CT_6/CT_5$. Therefore, it is favorable for avoiding the lens elements with an excessively thin thickness or an excessively thick thickness which are easily cracked or deformed during the manufacturing processes. In some embodiments, the following condition is satisfied: $2.0 < CT_6/CT_5 < 4.0$.

When a focal length of the fifth lens element is f_5 , and a focal length of the sixth lens element is f_6 , the following condition is satisfied: $-0.65 < f_6/f_5 < 0$. Therefore, it is favorable for balancing the refractive powers of the fifth lens element and the sixth lens element so as to reduce the back focal length of the image capturing lens assembly. In some embodiments, the following condition is satisfied: $-0.50 < f_6/f_5 < 0$.

When a curvature radius of the object-side surface of the sixth lens element is R_{11} , and a curvature radius of the image-side surface of the sixth lens element is R_{12} , the following condition is satisfied: $0 < (R_{11} + R_{12}) / (R_{11} - R_{12})$. Therefore, it is favorable for further correcting the astigmatism and reducing the back focal length thereof.

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When a focal length of the image capturing lens assembly is f , the focal length of the third lens element is f_3 , the focal length of the fourth lens element is f_4 , and the focal length of the fifth lens element is f_5 , the following condition is satisfied: $0.20 < |f/f_3| + |f/f_4| + |f/f_5| < 1.35$. Therefore, it is favorable for balancing the arrangement of the refractive powers so as to keep a compact size. In some embodiments, the following condition is satisfied: $0.20 < |f/f_3| + |f/f_4| + |f/f_5| < 1.0$.

When an Abbe number of the first lens element is V_1 , an Abbe number of the second lens element is V_2 , and an Abbe number of the fourth lens element is V_4 , the following condition is satisfied: $0.6 < (V_2 + V_4)/V_1 < 1.2$. Therefore, it is favorable for effectively correcting the chromatic aberration of the image capturing lens assembly.

When a curvature radius of the object-side surface of the third lens element is R_5 , and a curvature radius of the image-side surface of the third lens element is R_6 , the following condition is satisfied: $R_5/|R_6| < 1.0$. Therefore, it is favorable for correcting the astigmatism and spherical aberration of the image capturing lens assembly.

When the focal length of the image capturing lens assembly is f , the central thickness of the fourth lens element is CT_4 , the central thickness of the fifth lens element is CT_5 , and an axial distance between the fourth lens element and the fifth lens element is T_{45} , the following condition is satisfied: $5 \leq f/(CT_4 + T_{45} + CT_5) < 10$. Therefore, it is favorable for avoiding the lens elements which are easily crackled or deformed during the manufacturing processes and maintaining a compact size.

According to the image capturing lens assembly of the present disclosure, the lens elements thereof can be made of glass or plastic material. When the lens elements are made of glass material, the distribution of the refractive power of the image capturing lens assembly may be more flexible to design. When the lens elements are made of plastic material, the manufacturing cost can be effectively reduced. Furthermore, surfaces of each lens element can be arranged to be aspheric, since the aspheric surface of the lens element is easy to form a shape other than spherical surface so as to have more controllable variables for eliminating the aberration thereof, and to further decrease the required number of the lens elements. Therefore, the total track length of the image capturing lens assembly can also be reduced.

According to the image capturing lens assembly of the present disclosure, each of an object-side surface and an image-side surface has a paraxial region and an off-axis region. The paraxial region refers to the region of the surface where light rays travel close to the optical axis, and the off-axis region refers to the region of the surface where light rays travel away from the optical axis. Particularly, when the lens element has a convex surface, it indicates that the surface is convex in the paraxial region thereof; when the lens element has a concave surface, it indicates that the surface is concave in the paraxial region thereof.

According to the image capturing lens assembly of the present disclosure, the image capturing lens assembly can include at least one stop, such as an aperture stop, a glare stop or a field stop. Said glare stop or said field stop is for eliminating the stray light and thereby improving the image resolution thereof.

According to the image capturing lens assembly of the present disclosure, an aperture stop can be configured as a front stop or a middle stop. A front stop disposed between an imaged object and the first lens element can provide a longer distance between an exit pupil of the image capturing lens assembly and the image plane and thereby improves the image-sensing efficiency of an image sensor. A middle stop

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disposed between the first lens element and the image plane is favorable for enlarging the field of view of the image capturing lens assembly and thereby provides a wider field of view for the same.

The present image capturing lens assembly can be optionally applied to moving focus optical systems. According to the image capturing lens assembly of the present disclosure, the image capturing lens assembly is featured with good correction ability and high image quality, and can be applied to 3D (three-dimensional) image capturing applications, in products such as digital cameras, mobile devices, digital tablets, wearable devices and other mobile terminals.

According to the present disclosure, an image capturing device is provided. The image capturing device includes the image capturing lens assembly according to the aforementioned image capturing lens assembly of the present disclosure, and an image sensor, wherein the image sensor is disposed on an image plane of the aforementioned image capturing lens assembly. In some embodiments, the image capturing device can further include a barrel member, a holding member or a combination thereof.

In FIG. 17, FIG. 18 and FIG. 19, an imaging device 10 may be installed in but not limited to a mobile terminal, including a smart phone (FIG. 17), a tablet personal computer (FIG. 18) or a wearable device (FIG. 19). The three exemplary figures of different kinds of mobile terminal are only exemplary for showing the imaging device of present disclosure installing in a mobile terminal and is not limited thereto. In some embodiments, the mobile terminal can further include, but not limited to, display, a control unit, a random access memory unit (RAM), a read only memory unit (ROM) or a combination thereof.

According to the above description of the present disclosure, the following 1st-8th specific embodiments are provided for further explanation.

1st Embodiment

FIG. 1 is a schematic view of an image capturing device according to the 1st embodiment of the present disclosure. FIG. 2 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 1st embodiment. In FIG. 1, the image capturing device includes the image capturing lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor 190. The image capturing lens assembly includes, in order from an object side to an image side, an aperture stop 100, a first lens element 110, a second lens element 120, a third lens element 130, a fourth lens element 140, a fifth lens element 150, a sixth lens element 160, an IR-cut filter 170 and an image plane 180, wherein the image capturing lens assembly has a total of six lens elements (110-160) with refractive power.

The first lens element 110 with positive refractive power has an object-side surface 111 being convex in a paraxial region thereof and an image-side surface 112 being concave in a paraxial region thereof. The first lens element 110 is made of plastic material and has the object-side surface 111 and the image-side surface 112 being both aspheric.

The second lens element 120 with negative refractive power has an object-side surface 121 being convex in a paraxial region thereof and an image-side surface 122 being concave in a paraxial region thereof. The second lens element 120 is made of plastic material and has the object-side surface 121 and the image-side surface 122 being both aspheric.

The third lens element 130 with positive refractive power has an object-side surface 131 being convex in a paraxial

region thereof and an image-side surface **132** being convex in a paraxial region thereof. The third lens element **130** is made of plastic material and has the object-side surface **131** and the image-side surface **132** being both aspheric.

The fourth lens element **140** with negative refractive power has an object-side surface **141** being concave in a paraxial region thereof and an image-side surface **142** being convex in a paraxial region thereof. The fourth lens element **140** is made of plastic material and has the object-side surface **141** and the image-side surface **142** being both aspheric.

The fifth lens element **150** with negative refractive power has an object-side surface **151** being convex in a paraxial region thereof and an image-side surface **152** being concave in a paraxial region thereof, wherein the image-side surface **152** of the fifth lens element **150** has at least one convex shape in an off-axis region thereof. The fifth lens element **150** is made of plastic material and has the object-side surface **151** and the image-side surface **152** being both aspheric.

The sixth lens element **160** with negative refractive power has an object-side surface **161** being convex in a paraxial region thereof and an image-side surface **162** being concave in a paraxial region thereof, wherein the image-side surface **162** of the sixth lens element **160** has at least one convex shape in an off-axis region thereof. The sixth lens element **160** is made of plastic material and has the object-side surface **161** and the image-side surface **162** being both aspheric.

The IR-cut filter **170** is made of glass and located between the sixth lens element **160** and the image plane **180**, and will not affect the focal length of the image capturing lens assembly. The image sensor **190** is disposed on the image plane **180** of the image capturing lens assembly.

The equation of the aspheric surface profiles of the aforementioned lens elements of the 1st embodiment is expressed as follows:

$$X(Y) = (Y^2/R)/(1 + \sqrt{1 - (1+k) \times (Y/R)^2}) + \sum(A_i) \times (Y^i),$$

where,

X is the relative distance between a point on the aspheric surface spaced at a distance Y from the optical axis and the tangential plane at the aspheric surface vertex on the optical axis;

Y is the vertical distance from the point on the aspheric surface to the optical axis;

R is the curvature radius;

k is the conic coefficient; and

A_i is the i-th aspheric coefficient, and in the embodiments, i may be, but is not limited to, 4, 6, 8, 10, 12, 14 and 16.

In the image capturing lens assembly of the image capturing device according to the 1st embodiment, when a focal length of the image capturing lens assembly is f, an f-number of the image capturing lens assembly is Fno, and half of a maximal field of view of the image capturing lens assembly is HFOV, these parameters have the following values: f=4.81 mm; Fno=2.25; and HFOV=38.5 degrees.

In the image capturing lens assembly of the image capturing device according to the 1st embodiment, when an Abbe number of the first lens element **110** is V1, an Abbe number of

the second lens element **120** is V2, and an Abbe number of the fourth lens element **140** is V4, the following condition is satisfied: (V2+V4)/V1=0.83.

In the image capturing lens assembly of the image capturing device according to the 1st embodiment, when an axial distance between the third lens element **130** and the fourth lens element **140** is T34, and a central thickness of the fourth lens element **140** is CT4, the following condition is satisfied: T34/CT4=1.97.

In the image capturing lens assembly of the image capturing device according to the 1st embodiment, when a central thickness of the fifth lens element **150** is CT5, and a central thickness of the sixth lens element **160** is CT6, the following condition is satisfied: CT6/CT5=2.46.

In the image capturing lens assembly of the image capturing device according to the 1st embodiment, when the focal length of the image capturing lens assembly is f, the central thickness of the fourth lens element **140** is CT4, the central thickness of the fifth lens element **150** is CT5, and an axial distance between the fourth lens element **140** and the fifth lens element **150** is T45, the following condition is satisfied: f/(CT4+T45+CT5)=6.66.

In the image capturing lens assembly of the image capturing device according to the 1st embodiment, when a curvature radius of the object-side surface **131** of the third lens element **130** is R5, and a curvature radius of the image-side surface **132** of the third lens element **130** is R6, the following condition is satisfied: R5/|R6|=0.44.

In the image capturing lens assembly of the image capturing device according to the 1st embodiment, when a curvature radius of the object-side surface **161** of the sixth lens element **160** is R11, and a curvature radius of the image-side surface **162** of the sixth lens element **160** is R12, the following condition is satisfied: (R11+R12)/(R11-R12)=2.83.

In the image capturing lens assembly of the image capturing device according to the 1st embodiment, when a focal length of the third lens element **130** is f3, and a focal length of the fourth lens element **140** is f4, the following condition is satisfied: |f3/f4|=0.08.

In the image capturing lens assembly of the image capturing device according to the 1st embodiment, when the focal length of the image capturing lens assembly is f, the focal length of the third lens element **130** is f3, the focal length of the fourth lens element **140** is f4, and a focal length of the fifth lens element **150** is f5, the following condition is satisfied: |f/f3|+|f/f4|+|f/f5|=0.45.

In the image capturing lens assembly of the image capturing device according to the 1st embodiment, when the focal length of the fifth lens element **150** is f5, and a focal length of the sixth lens element **160** is f6, the following condition is satisfied: f6/|f5|=−0.05.

The detailed optical data of the 1st embodiment are shown in Table 1 and the aspheric surface data are shown in Table 2 below.

TABLE 1

1st Embodiment						
f = 4.81 mm, Fno = 2.25, HFOV = 38.5 deg.						
Surface #		Curvature Radius	Thickness	Material	Index	Abbe # Focal Length
0	Object	Plano	Infinity			
1	Ape. Stop	Plano	−0.377			

TABLE 1-continued

1st Embodiment f = 4.81 mm, Fno = 2.25, HFOV = 38.5 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
2	Lens 1	1.739	ASP	0.571	Plastic	1.544	55.9	3.64
3		12.479	ASP	0.088				
4	Lens 2	6.063	ASP	0.250	Plastic	1.640	23.3	-7.07
5		2.549	ASP	0.376				
6	Lens 3	9.732	ASP	0.526	Plastic	1.544	55.9	12.47
7		-21.981	ASP	0.591				
8	Lens 4	-3.462	ASP	0.300	Plastic	1.640	23.3	-160.43
9		-3.704	ASP	0.033				
10	Lens 5	4.260	ASP	0.389	Plastic	1.544	55.9	-160.26
11		3.931	ASP	0.376				
12	Lens 6	4.360	ASP	0.958	Plastic	1.535	55.7	-8.72
13		2.081	ASP	0.500				
14	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—	
15		Plano	0.303					
16	Image	Plano	—					

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 2

Aspheric Coefficients						
Surface #	2	3	4	5	6	7
k =	3.3268E-01	-8.9766E+01	-6.3991E+01	-2.3334E+01	-9.0000E+01	-1.0000E+00
A4 =	-1.0713E-02	-1.0027E-01	-1.7198E-01	2.9618E-02	-2.9658E-02	-5.1583E-02
A6 =	2.4708E-02	2.7670E-01	4.2971E-01	1.0865E-01	-4.9890E-02	7.3964E-02
A8 =	-4.9234E-02	-4.3252E-01	-6.3398E-01	-2.2847E-01	1.6187E-01	-2.1054E-01
A10 =	4.8851E-02	4.3838E-01	6.2636E-01	3.3086E-01	-2.2919E-01	3.0593E-01
A12 =	-2.1055E-02	-2.4311E-01	-3.5726E-01	-2.3798E-01	1.6656E-01	-2.4596E-01
A14 =	3.0175E-03	5.2463E-02	8.1977E-02	6.8551E-02	-4.3715E-02	1.0307E-01
A16 =						-1.6286E-02
Surface #	8	9	10	11	12	13
k =	1.0000E+00	-8.3240E+00	-6.8947E+01	-3.0724E+01	-2.3613E+00	-1.0632E+00
A4 =	-1.1521E-02	-5.1333E-02	5.0979E-02	-3.1368E-03	-1.6739E-01	-1.2698E-01
A6 =	4.9413E-02	6.9357E-02	-8.6159E-02	9.5694E-05	6.0303E-02	4.3455E-02
A8 =	-7.0575E-02	-6.8042E-02	4.8745E-02	-1.0284E-02	-1.1751E-02	-1.1887E-02
A10 =	2.7030E-02	2.4228E-02	-2.6304E-02	4.1328E-03	1.4612E-03	2.1387E-03
A12 =	-4.7650E-03	-2.7416E-03	7.1987E-03	-5.9196E-04	-1.1910E-04	-2.3292E-04
A14 =			-6.7221E-04	2.8822E-05	5.9063E-06	1.3765E-05
A16 =					-1.3574E-07	-3.3618E-07

In Table 1, the curvature radius, the thickness and the focal length are shown in millimeters (mm). Surface numbers 0-16 represent the surfaces sequentially arranged from the object-side to the image-side along the optical axis. In Table 2, k represents the conic coefficient of the equation of the aspheric surface profiles. A4-A16 represent the aspheric coefficients ranging from the 4th order to the 16th order. The tables presented below for each embodiment are the corresponding schematic parameter and aberration curves, and the definitions of the tables are the same as Table 1 and Table 2 of the 1st embodiment. Therefore, an explanation in this regard will not be provided again.

2nd Embodiment

FIG. 3 is a schematic view of an image capturing device according to the 2nd embodiment of the present disclosure. FIG. 4 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 2nd embodiment. In FIG. 3, the image capturing device includes the image cap-

turing lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor **290**. The image capturing lens assembly includes, in order from an object side to an image side, an aperture stop **200**, a first lens element **210**, a second lens element **220**, a third lens element **230**, a fourth lens element **240**, a fifth lens element **250**, a sixth lens element **260**, an IR-cut filter **270** and an image plane **280**, wherein the image capturing lens assembly has a total of six lens elements (**210-260**) with refractive power.

The first lens element **210** with positive refractive power has an object-side surface **211** being convex in a paraxial region thereof and an image-side surface **212** being concave in a paraxial region thereof. The first lens element **210** is made of plastic material and has the object-side surface **211** and the image-side surface **212** being both aspheric.

The second lens element **220** with negative refractive power has an object-side surface **221** being convex in a paraxial region thereof and an image-side surface **222** being concave in a paraxial region thereof. The second lens element **220** is made of plastic material and has the object-side surface **221** and the image-side surface **222** being both aspheric.

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The third lens element **230** with positive refractive power has an object-side surface **231** being convex in a paraxial region thereof and an image-side surface **232** being convex in a paraxial region thereof. The third lens element **230** is made of plastic material and has the object-side surface **231** and the image-side surface **232** being both aspheric.

The fourth lens element **240** with negative refractive power has an object-side surface **241** being concave in a paraxial region thereof and an image-side surface **242** being convex in a paraxial region thereof. The fourth lens element **240** is made of plastic material and has the object-side surface **241** and the image-side surface **242** being both aspheric.

The fifth lens element **250** with negative refractive power has an object-side surface **251** being convex in a paraxial region thereof and an image-side surface **252** being concave in a paraxial region thereof, wherein the image-side surface **252** of the fifth lens element **250** has at least one convex shape in an off-axis region thereof. The fifth lens element **250** is

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made of plastic material and has the object-side surface **251** and the image-side surface **252** being both aspheric.

The sixth lens element **260** with negative refractive power has an object-side surface **261** being convex in a paraxial region thereof and an image-side surface **262** being concave in a paraxial region thereof, wherein the image-side surface **262** of the sixth lens element **260** has at least one convex shape in an off-axis region thereof. The sixth lens element **260** is made of plastic material and has the object-side surface **261** and the image-side surface **262** being both aspheric.

The IR-cut filter **270** is made of glass and located between the sixth lens element **260** and the image plane **280**, and will not affect the focal length of the image capturing lens assembly. The image sensor **290** is disposed on the image plane **280** of the image capturing lens assembly.

The detailed optical data of the 2nd embodiment are shown in Table 3 and the aspheric surface data are shown in Table 4 below.

TABLE 3

2nd Embodiment								
f = 4.77 mm, Fno = 2.25, HFOV = 38.5 deg.								
Surface #		Curvature	Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano		Infinity				
1	Ape. Stop	Plano		-0.365				
2	Lens 1	1.724	ASP	0.548	Plastic	1.544	55.9	3.94
3		7.789	ASP	0.121				
4	Lens 2	5.221	ASP	0.250	Plastic	1.640	23.3	-7.58
5		2.467	ASP	0.355				
6	Lens 3	5.802	ASP	0.394	Plastic	1.544	55.9	10.23
7		-133.194	ASP	0.567				
8	Lens 4	-1.965	ASP	0.320	Plastic	1.640	23.3	-45.47
9		-2.241	ASP	0.035				
10	Lens 5	5.063	ASP	0.350	Plastic	1.544	55.9	-101.76
11		4.526	ASP	0.321				
12	Lens 6	2.523	ASP	0.844	Plastic	1.535	55.7	-19.96
13		1.803	ASP	0.700				
14	IR-cut filter	Plano		0.210	Glass	1.517	64.2	—
15		Plano		0.456				
16	Image	Plano		—				

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 4

Aspheric Coefficients						
Surface #	2	3	4	5	6	7
k =	7.4053E-02	-8.9766E+01	-6.4009E+01	-2.3331E+01	-9.0000E+01	-1.0000E+00
A4 =	-1.4685E-02	-7.4217E-02	-1.6195E-01	2.9806E-02	1.8500E-02	-3.7499E-02
A6 =	6.3750E-02	1.6579E-01	3.1434E-01	2.9755E-02	-1.0342E-01	1.1287E-01
A8 =	-1.3644E-01	-2.4105E-01	-3.4850E-01	2.8682E-02	2.0664E-01	-3.2933E-01
A10 =	1.5040E-01	2.6511E-01	3.1316E-01	-8.2191E-03	-2.4160E-01	4.8453E-01
A12 =	-7.8048E-02	-1.6519E-01	-1.9231E-01	-2.3842E-02	1.5529E-01	-3.9970E-01
A14 =	1.4466E-02	3.8651E-02	4.9420E-02	1.3610E-02	-3.7207E-02	1.7015E-01
A16 =						-2.7247E-02
Surface #	8	9	10	11	12	13
k =	6.3665E-02	-8.3513E+00	-6.8947E+01	-3.0359E+01	-2.6205E+00	-1.1289E+00
A4 =	3.1609E-03	-9.0390E-02	1.0855E-01	4.9392E-04	-2.1406E-01	-1.6736E-01
A6 =	1.2794E-01	9.4024E-02	-1.5034E-01	1.0722E-02	7.8713E-02	6.3187E-02
A8 =	-1.7280E-01	-8.4174E-02	9.7160E-02	-1.9050E-02	-1.5482E-02	-1.8634E-02
A10 =	9.1142E-02	3.4384E-02	-4.7262E-02	6.6204E-03	1.9661E-03	3.6106E-03
A12 =	-2.0911E-02	-4.6602E-03	1.1788E-02	-8.9634E-04	-1.7349E-04	-4.2366E-04
A14 =			-1.0698E-03	4.1311E-05	1.0207E-05	2.7192E-05
A16 =					-2.9673E-07	-7.2770E-07

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In the 2nd embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 2nd embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 3 and Table 4 as the following values and satisfy the following conditions:

2nd Embodiment			
f [mm]	4.77	$f/(CT4 + T45 + CT5)$	6.77
Fno	2.25	$R5/ R6 $	0.04
HFOV [deg.]	38.5	$(R11 + R12)/(R11 - R12)$	6.01
$(V2 + V4)/V1$	0.83	$ f3/f4 $	0.22
T34/CT4	1.77	$ f/f3 + f/f4 + f/f5 $	0.62
CT6/CT5	2.41	$f6/ f5 $	-0.20

3rd Embodiment

FIG. 5 is a schematic view of an image capturing device according to the 3rd embodiment of the present disclosure. FIG. 6 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 3rd embodiment. In FIG. 5, the image capturing device includes the image capturing lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor 390. The image capturing lens assembly includes, in order from an object side to an image side, an aperture stop 300, a first lens element 310, a second lens element 320, a third lens element 330, a fourth lens element 340, a fifth lens element 350, a sixth lens element 360, an IR-cut filter 370 and an image plane 380, wherein the image capturing lens assembly has a total of six lens elements (310-360) with refractive power.

The first lens element 310 with positive refractive power has an object-side surface 311 being convex in a paraxial region thereof and an image-side surface 312 being concave in a paraxial region thereof. The first lens element 310 is made of plastic material and has the object-side surface 311 and the image-side surface 312 being both aspheric.

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The second lens element 320 with negative refractive power has an object-side surface 321 being convex in a paraxial region thereof and an image-side surface 322 being concave in a paraxial region thereof. The second lens element 320 is made of plastic material and has the object-side surface 321 and the image-side surface 322 being both aspheric.

The third lens element 330 with positive refractive power has an object-side surface 331 being convex in a paraxial region thereof and an image-side surface 332 being concave in a paraxial region thereof. The third lens element 330 is made of plastic material and has the object-side surface 331 and the image-side surface 332 being both aspheric.

The fourth lens element 340 with negative refractive power has an object-side surface 341 being concave in a paraxial region thereof and an image-side surface 342 being convex in a paraxial region thereof. The fourth lens element 340 is made of plastic material and has the object-side surface 341 and the image-side surface 342 being both aspheric.

The fifth lens element 350 with positive refractive power has an object-side surface 351 being convex in a paraxial region thereof and an image-side surface 352 being concave in a paraxial region thereof, wherein the image-side surface 352 of the fifth lens element 350 has at least one convex shape in an off-axis region thereof. The fifth lens element 350 is made of plastic material and has the object-side surface 351 and the image-side surface 352 being both aspheric.

The sixth lens element 360 with negative refractive power has an object-side surface 361 being convex in a paraxial region thereof and an image-side surface 362 being concave in a paraxial region thereof, wherein the image-side surface 362 of the sixth lens element 360 has at least one convex shape in an off-axis region thereof. The sixth lens element 360 is made of plastic material and has the object-side surface 361 and the image-side surface 362 being both aspheric.

The IR-cut filter 370 is made of glass and located between the sixth lens element 360 and the image plane 380, and will not affect the focal length of the image capturing lens assembly. The image sensor 390 is disposed on the image plane 380 of the image capturing lens assembly.

The detailed optical data of the 3rd embodiment are shown in Table 5 and the aspheric surface data are shown in Table 6 below.

TABLE 5

3rd Embodiment								
f = 4.70 mm, Fno = 2.25, HFOV = 39.0 deg.								
Surface #		Curvature Radius		Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano		Infinity				
1	Ape. Stop	Plano		-0.366				
2	Lens 1	1.667	ASP	0.538	Plastic	1.544	55.9	4.15
3		5.641	ASP	0.103				
4	Lens 2	4.928	ASP	0.258	Plastic	1.640	23.3	-8.34
5		2.509	ASP	0.346				
6	Lens 3	4.289	ASP	0.330	Plastic	1.544	55.9	11.16
7		14.206	ASP	0.572				
8	Lens 4	-1.859	ASP	0.306	Plastic	1.640	23.3	-104.46
9		-2.035	ASP	0.088				
10	Lens 5	4.913	ASP	0.320	Plastic	1.544	55.9	132.18
11		5.152	ASP	0.401				
12	Lens 6	2.172	ASP	0.672	Plastic	1.544	55.9	-16.88
13		1.565	ASP	0.300				

TABLE 5-continued

3rd Embodiment f = 4.70 mm, Fno = 2.25, HFOV = 39.0 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
14	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—
15		Plano	0.895				
16	Image	Plano	—				

Note:

Reference wavelength is 587.6 nm (d-line).

The effective radius of Surface 7 is 1.200 mm.

TABLE 6

Aspheric Coefficients						
Surface #	2	3	4	5	6	7
k =	-8.4038E-02	-8.9547E+01	-6.4010E+01	-2.4285E+01	-8.6803E+01	-1.0000E+00
A4 =	-7.8703E-03	-8.2459E-02	-2.0405E-01	3.7770E-03	9.2853E-02	-2.5871E-02
A6 =	5.9913E-02	1.5282E-01	3.5738E-01	5.8324E-02	-3.2692E-01	8.3723E-02
A8 =	-1.3677E-01	-1.7751E-01	-2.8942E-01	9.6780E-02	5.7568E-01	-3.2882E-01
A10 =	1.6183E-01	1.9493E-01	1.8166E-01	-1.6627E-01	-6.1125E-01	5.3108E-01
A12 =	-8.9070E-02	-1.3295E-01	-9.9510E-02	1.0248E-01	3.5382E-01	-4.5932E-01
A14 =	1.7156E-02	3.3292E-02	2.7323E-02	-2.0465E-02	-8.0698E-02	2.0067E-01
A16 =						-3.2593E-02
Surface #	8	9	10	11	12	13
k =	4.4489E-01	-7.9242E+00	-5.8707E+01	-3.1539E+01	-2.2440E+00	-1.3186E+00
A4 =	4.6495E-02	-5.0830E-02	1.3369E-01	6.6389E-03	-2.0424E-01	-1.8625E-01
A6 =	1.0145E-01	4.7972E-02	-1.4422E-01	2.3454E-02	5.0691E-02	6.8738E-02
A8 =	-1.9765E-01	-7.1100E-02	8.4735E-02	-2.5929E-02	-1.3122E-03	-1.9279E-02
A10 =	1.3092E-01	3.8206E-02	-3.6039E-02	7.8158E-03	-1.4728E-03	3.5603E-03
A12 =	-3.4475E-02	-6.2746E-03	7.7999E-03	-9.7860E-04	2.7912E-04	-4.0286E-04
A14 =			-6.2196E-04	4.2677E-05	-2.0899E-05	2.5503E-05
A16 =					5.8005E-07	-6.8953E-07

In the 3rd embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 3rd embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 5 and Table 6 as the following values and satisfy the following conditions:

3rd Embodiment			
f [mm]	4.70	$f/(CT4 + T45 + CT5)$	6.58
Fno	2.25	$R5/ R6 $	0.30
HFOV [deg.]	39.0	$(R11 + R12)/(R11 - R12)$	6.16
$(V2 + V4)/V1$	0.83	$ f3/f4 $	0.11
T34/CT4	1.87	$ f7/f3 + f7/f4 + f7/f5 $	0.50
CT6/CT5	2.10	$f6/f5$	-0.13

4th Embodiment

FIG. 7 is a schematic view of an image capturing device according to the 4th embodiment of the present disclosure. FIG. 8 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 4th embodiment. In FIG. 7, the image capturing device includes the image capturing lens assembly (not otherwise herein labeled) of the

present disclosure and an image sensor 490. The image capturing lens assembly includes, in order from an object side to an image side, an aperture stop 400, a first lens element 410, a second lens element 420, a third lens element 430, a fourth lens element 440, a fifth lens element 450, a sixth lens element 460, an IR-cut filter 470 and an image plane 480, wherein the image capturing lens assembly has a total of six lens elements (410-460) with refractive power.

The first lens element 410 with positive refractive power has an object-side surface 411 being convex in a paraxial region thereof and an image-side surface 412 being concave in a paraxial region thereof. The first lens element 410 is made of plastic material and has the object-side surface 411 and the image-side surface 412 being both aspheric.

The second lens element 420 with negative refractive power has an object-side surface 421 being convex in a paraxial region thereof and an image-side surface 422 being concave in a paraxial region thereof. The second lens element 420 is made of plastic material and has the object-side surface 421 and the image-side surface 422 being both aspheric.

The third lens element 430 with positive refractive power has an object-side surface 431 being convex in a paraxial region thereof and an image-side surface 432 being concave in a paraxial region thereof. The third lens element 430 is made of plastic material and has the object-side surface 431 and the image-side surface 432 being both aspheric.

The fourth lens element 440 with negative refractive power has an object-side surface 441 being concave in a paraxial region thereof and an image-side surface 442 being convex in a paraxial region thereof. The fourth lens element 440 is made

of plastic material and has the object-side surface **441** and the image-side surface **442** being both aspheric.

The fifth lens element **450** with negative refractive power has an object-side surface **451** being convex in a paraxial region thereof and an image-side surface **452** being concave in a paraxial region thereof, wherein the image-side surface **452** of the fifth lens element **450** has at least one convex shape in an off-axis region thereof. The fifth lens element **450** is

The IR-cut filter **470** is made of glass and located between the sixth lens element **460** and the image plane **480**, and will not affect the focal length of the image capturing lens assembly. The image sensor **490** is disposed on the image plane **480** of the image capturing lens assembly.

The detailed optical data of the 4th embodiment are shown in Table 7 and the aspheric surface data are shown in Table 8 below.

TABLE 7

4th Embodiment								
f = 4.67 mm, Fno = 2.10, HFOV = 39.0 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.390					
2	Lens 1	1.771	ASP	0.609	Plastic	1.544	55.9	4.12
3		7.420	ASP	0.164				
4	Lens 2	4.452	ASP	0.250	Plastic	1.650	21.4	-7.54
5		2.281	ASP	0.255				
6	Lens 3	4.732	ASP	0.370	Plastic	1.544	55.9	9.11
7		100.000	ASP	0.570				
8	Lens 4	-2.172	ASP	0.370	Plastic	1.633	23.4	-21.12
9		-2.765	ASP	0.035				
10	Lens 5	9.786	ASP	0.529	Plastic	1.583	30.2	-47.57
11		7.091	ASP	0.162				
12	Lens 6	2.108	ASP	0.942	Plastic	1.530	55.8	155.04
13		1.829	ASP	0.300				
14	IR-cut filter	Plano	0.175	Glass	1.517	64.2	—	
15		Plano	0.827					
16	Image	Plano	—					

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 8

Aspheric Coefficients						
Surface #	2	3	4	5	6	7
k =	5.5737E-02	-2.7413E+01	-4.8329E+01	-2.2304E+01	-7.1506E+01	5.0000E+00
A4 =	-1.7734E-02	-6.9981E-02	-1.4885E-01	1.8021E-02	2.0733E-02	-3.7949E-02
A6 =	6.7512E-02	1.6293E-01	3.0893E-01	3.5194E-02	-1.1080E-01	1.1238E-01
A8 =	-1.3622E-01	-2.4485E-01	-3.5776E-01	2.0474E-02	2.0870E-01	-3.2989E-01
A10 =	1.4639E-01	2.6456E-01	3.1330E-01	-1.0625E-02	-2.3725E-01	4.8808E-01
A12 =	-7.5716E-02	-1.6312E-01	-1.9005E-01	-2.2291E-02	1.5580E-01	-3.9872E-01
A14 =	1.4466E-02	3.8651E-02	4.9420E-02	1.3610E-02	-3.8007E-02	1.7064E-01
A16 =	-7.2878E-12	1.8789E-12	-5.7321E-13	2.0027E-14	9.8288E-13	-2.7247E-02
Surface #	8	9	10	11	12	13
k =	3.6640E-01	-9.7808E+00	-1.0000E+00	-4.5335E+00	-1.7869E+00	-1.6512E+00
A4 =	-2.7174E-02	-6.9181E-02	1.3140E-01	3.9838E-02	-1.9951E-01	-1.3523E-01
A6 =	1.4337E-01	-1.4195E-02	-2.6320E-01	-5.8872E-02	7.6619E-02	4.8563E-02
A8 =	-1.7616E-01	5.5231E-02	2.2661E-01	2.4225E-02	-1.9175E-02	-1.3625E-02
A10 =	9.5041E-02	-6.2322E-02	-1.3631E-01	-6.0428E-03	3.4621E-03	2.5769E-03
A12 =	-2.0543E-02	3.6447E-02	5.1427E-02	8.3122E-04	-4.1374E-04	-2.9894E-04
A14 =	1.9418E-11	-1.0063E-02	-1.1107E-02	-4.2307E-05	2.8315E-05	1.9036E-05
A16 =	3.9859E-12	1.0139E-03	1.0450E-03	-7.1756E-07	-8.2973E-07	-5.0562E-07

made of plastic material and has the object-side surface **451** and the image-side surface **452** being both aspheric.

The sixth lens element **460** with positive refractive power has an object-side surface **461** being convex in a paraxial region thereof and an image-side surface **462** being concave in a paraxial region thereof, wherein the image-side surface **462** of the sixth lens element **460** has at least one convex shape in an off-axis region thereof. The sixth lens element **460** is made of plastic material and has the object-side surface **461** and the image-side surface **462** being both aspheric.

In the 4th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 4th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 7 and Table 8 as the following values and satisfy the following conditions:

4th Embodiment			
f [mm]	4.67	$f/(CT4 + T45 + CT5)$	5.00
Fno	2.10	$R5/ R6 $	0.05
HFOV [deg.]	39.0	$(R11 + R12)/(R11 - R12)$	14.09
$(V2 + V4)/V1$	0.80	$ f3/f4 $	0.43
T34/CT4	1.54	$ f/f3 + f/f4 + f/f5 $	0.83
CT6/CT5	1.78	$f6/ f5 $	3.26

5th Embodiment

FIG. 9 is a schematic view of an image capturing device according to the 5th embodiment of the present disclosure. FIG. 10 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 5th embodiment. In FIG. 9, the image capturing device includes the image capturing lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor 590. The image capturing lens assembly includes, in order from an object side to an image side, an aperture stop 500, a first lens element 510, a second lens element 520, a third lens element 530, a fourth lens element 540, a fifth lens element 550, a sixth lens element 560, an IR-cut filter 570 and an image plane 580, wherein the image capturing lens assembly has a total of six lens elements (510-560) with refractive power.

The first lens element 510 with positive refractive power has an object-side surface 511 being convex in a paraxial region thereof and an image-side surface 512 being concave in a paraxial region thereof. The first lens element 510 is made of glass material and has the object-side surface 511 and the image-side surface 512 being both aspheric.

The second lens element 520 with negative refractive power has an object-side surface 521 being convex in a paraxial region thereof and an image-side surface 522 being concave in a paraxial region thereof. The second lens element

520 is made of plastic material and has the object-side surface 521 and the image-side surface 522 being both aspheric.

The third lens element 530 with positive refractive power has an object-side surface 531 being convex in a paraxial region thereof and an image-side surface 532 being convex in a paraxial region thereof. The third lens element 530 is made of plastic material and has the object-side surface 531 and the image-side surface 532 being both aspheric.

The fourth lens element 540 with negative refractive power has an object-side surface 541 being concave in a paraxial region thereof and an image-side surface 542 being convex in a paraxial region thereof. The fourth lens element 540 is made of plastic material and has the object-side surface 541 and the image-side surface 542 being both aspheric.

The fifth lens element 550 with positive refractive power has an object-side surface 551 being convex in a paraxial region thereof and an image-side surface 552 being concave in a paraxial region thereof, wherein the image-side surface 552 of the fifth lens element 550 has at least one convex shape in an off-axis region thereof. The fifth lens element 550 is made of plastic material and has the object-side surface 551 and the image-side surface 552 being both aspheric.

The sixth lens element 560 with negative refractive power has an object-side surface 561 being convex in a paraxial region thereof and an image-side surface 562 being concave in a paraxial region thereof, wherein the image-side surface 562 of the sixth lens element 560 has at least one convex shape in an off-axis region thereof. The sixth lens element 560 is made of plastic material and has the object-side surface 561 and the image-side surface 562 being both aspheric.

The IR-cut filter 570 is made of glass and located between the sixth lens element 560 and the image plane 580, and will not affect the focal length of the image capturing lens assembly. The image sensor 590 is disposed on the image plane 580 of the image capturing lens assembly.

The detailed optical data of the 5th embodiment are shown in Table 9 and the aspheric surface data are shown in Table 10 below.

TABLE 9

5th Embodiment								
f = 4.65 mm, Fno = 2.00, HFOV = 39.4 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.438					
2	Lens 1	1.776 ASP	0.698	Glass	1.542	62.9	4.10	
3		7.240 ASP	0.163					
4	Lens 2	4.605 ASP	0.245	Plastic	1.639	23.5	-7.20	
5		2.268 ASP	0.212					
6	Lens 3	5.485 ASP	0.405	Plastic	1.544	55.9	7.80	
7		-18.310 ASP	0.573					
8	Lens 4	-2.466 ASP	0.386	Plastic	1.639	23.5	-24.70	
9		-3.099 ASP	0.035					
10	Lens 5	4.113 ASP	0.459	Plastic	1.544	55.9	39.40	
11		4.889 ASP	0.317					
12	Lens 6	4.358 ASP	1.008	Plastic	1.530	55.8	-9.80	
13		2.185 ASP	0.500					
14	IR-cut filter	Plano	0.175	Glass	1.517	64.2	—	
15		Plano	0.328					
16	Image	Plano	—					

Note:

Reference wavelength is 587.6 nm (d-line).

The effective radius of Surface 9 is 1.600 mm.

TABLE 10

Aspheric Coefficients						
Surface #	2	3	4	5	6	7
k =	7.2761E-02	-7.8709E+00	-4.3975E+01	-2.0942E+01	-9.0000E+01	-9.8779E-01
A4 =	-1.8792E-02	-6.7826E-02	-1.5644E-01	1.2450E-02	2.2177E-02	-3.7969E-02
A6 =	7.0183E-02	1.5916E-01	3.0362E-01	3.0894E-02	-1.0398E-01	1.1866E-01
A8 =	-1.3704E-01	-2.4339E-01	-3.5426E-01	2.0220E-02	2.0938E-01	-3.3050E-01
A10 =	1.4250E-01	2.6821E-01	3.1586E-01	-9.9047E-03	-2.3703E-01	4.8801E-01
A12 =	-7.2227E-02	-1.6340E-01	-1.9279E-01	-2.3967E-02	1.5768E-01	-3.9850E-01
A14 =	1.4466E-02	3.8651E-02	4.9420E-02	1.3610E-02	-3.9211E-02	1.7247E-01
A16 =	-3.2621E-12	-2.8109E-12	3.0701E-12	-1.5630E-12	1.3560E-12	-2.7247E-02
Surface #	8	9	10	11	12	13
k =	5.8669E-01	-6.4295E+00	-6.2065E+00	-6.8146E+00	-1.3274E+01	-1.0152E+00
A4 =	-3.6309E-02	-7.1717E-02	3.9040E-02	4.4305E-02	-1.2423E-01	-1.1578E-01
A6 =	1.3439E-01	3.7249E-02	-1.2057E-01	-7.7476E-02	3.7016E-02	3.5804E-02
A8 =	-1.6885E-01	1.2309E-04	9.4886E-02	4.0396E-02	-6.1136E-03	-9.2878E-03
A10 =	9.4794E-02	-2.7580E-02	-5.4978E-02	-1.3587E-02	8.5295E-04	1.6343E-03
A12 =	-2.0543E-02	2.4982E-02	2.1105E-02	2.8730E-03	-1.0364E-04	-1.7368E-04
A14 =	5.2088E-11	-8.5084E-03	-4.8362E-03	-3.5942E-04	7.9884E-06	9.9754E-06
A16 =	6.0014E-12	9.9912E-04	4.8374E-04	2.0549E-05	-2.6169E-07	-2.3709E-07

In the 5th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 5th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 9 and Table 10 as the following values and satisfy the following conditions:

5th Embodiment			
f [mm]	4.65	f/(CT4 + T45 + CT5)	5.28
Fno	2.00	R5/ R6	0.30
HFOV [deg.]	39.4	(R11 + R12)/(R11 - R12)	3.01
(V2 + V4)/V1	0.75	f3/f4	0.32
T34/CT4	1.48	f7/f3 + f7/f4 + f7/f5	0.90
CT6/CT5	2.20	f6/ f5	-0.25

6th Embodiment

FIG. 11 is a schematic view of an image capturing device according to the 6th embodiment of the present disclosure. FIG. 12 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 6th embodiment. In FIG. 11, the image capturing device includes the image capturing lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor 690. The image capturing lens assembly includes, in order from an object side to an image side, an aperture stop 600, a first lens element 610, a second lens element 620, a third lens element 630, a fourth lens element 640, a fifth lens element 650, a sixth lens element 660, an IR-cut filter 670 and an image plane 680, wherein the image capturing lens assembly has a total of six lens elements (610-660) with refractive power.

The first lens element 610 with positive refractive power has an object-side surface 611 being convex in a paraxial region thereof and an image-side surface 612 being convex in a paraxial region thereof. The first lens element 610 is made of

plastic material and has the object-side surface 611 and the image-side surface 612 being both aspheric.

The second lens element 620 with negative refractive power has an object-side surface 621 being convex in a paraxial region thereof and an image-side surface 622 being concave in a paraxial region thereof. The second lens element 620 is made of plastic material and has the object-side surface 621 and the image-side surface 622 being both aspheric.

The third lens element 630 with positive refractive power has an object-side surface 631 being convex in a paraxial region thereof and an image-side surface 632 being planar in a paraxial region thereof. The third lens element 630 is made of plastic material and has the object-side surface 631 and the image-side surface 632 being both aspheric.

The fourth lens element 640 with positive refractive power has an object-side surface 641 being concave in a paraxial region thereof and an image-side surface 642 being convex in a paraxial region thereof. The fourth lens element 640 is made of plastic material and has the object-side surface 641 and the image-side surface 642 being both aspheric.

The fifth lens element 650 with negative refractive power has an object-side surface 651 being convex in a paraxial region thereof and an image-side surface 652 being concave in a paraxial region thereof, wherein the image-side surface 652 of the fifth lens element 650 has at least one convex shape in an off-axis region thereof. The fifth lens element 650 is made of plastic material and has the object-side surface 651 and the image-side surface 652 being both aspheric.

The sixth lens element 660 with negative refractive power has an object-side surface 661 being convex in a paraxial region thereof and an image-side surface 662 being concave in a paraxial region thereof, wherein the image-side surface 662 of the sixth lens element 660 has at least one convex shape in an off-axis region thereof. The sixth lens element 660 is made of plastic material and has the object-side surface 661 and the image-side surface 662 being both aspheric.

The IR-cut filter 670 is made of glass and located between the sixth lens element 660 and the image plane 680, and will not affect the focal length of the image capturing lens assembly. The image sensor 690 is disposed on the image plane 680 of the image capturing lens assembly.

The detailed optical data of the 6th embodiment are shown in Table 11 and the aspheric surface data are shown in Table 12 below.

TABLE 11

6th Embodiment								
f = 5.30 mm, Fno = 2.32, HFOV = 36.0 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.350					
2	Lens 1	1.985 ASP	0.679	Plastic	1.544	55.9	3.58	
3		-97.576 ASP	0.095					
4	Lens 2	8.714 ASP	0.269	Plastic	1.634	23.8	-5.95	
5		2.601 ASP	0.428					
6	Lens 3	7.829 ASP	0.715	Plastic	1.544	55.9	14.39	
7		∞ ASP	0.559					
8	Lens 4	-2.965 ASP	0.251	Plastic	1.634	23.8	50.21	
9		-2.802 ASP	0.035					
10	Lens 5	2.955 ASP	0.381	Plastic	1.544	55.9	-23.76	
11		2.296 ASP	0.455					
12	Lens 6	5.782 ASP	1.163	Plastic	1.530	55.8	-11.15	
13		2.719 ASP	0.300					
14	IR-cut filter	Plano	0.300	Glass	1.517	64.2	—	
15		Plano	0.471					
16	Image	Plano	—					

Note:

Reference wavelength is 587.6 nm (d-line).

The effective radius of Surface 8 is 1.480 mm.

TABLE 12

Aspheric Coefficients						
Surface #	2	3	4	5	6	7
k =	1.2768E-01	-1.0000E+00	-4.0077E+01	-1.7837E+01	-7.2368E+01	-1.0000E+00
A4 =	-8.3803E-03	-3.9349E-02	-1.1695E-01	1.2503E-02	-1.7641E-02	-2.7393E-02
A6 =	3.3163E-02	9.6759E-02	1.8471E-01	1.5127E-02	1.6113E-02	4.3108E-02
A8 =	-6.7474E-02	-1.3155E-01	-1.8452E-01	2.7988E-03	-6.1359E-02	-7.4725E-02
A10 =	6.4743E-02	1.1182E-01	1.3424E-01	-7.8253E-04	9.7502E-02	6.1295E-02
A12 =	-2.9231E-02	-5.7033E-02	-6.5387E-02	-3.8154E-03	-8.3246E-02	-2.4982E-02
A14 =	4.0844E-03	1.1184E-02	1.4496E-02	3.6849E-03	3.8859E-02	4.0233E-03
A16 =	-1.0446E-04	-1.4710E-05	1.1523E-04	-9.4458E-13	-7.0622E-03	2.2135E-04
Surface #	8	9	10	11	12	13
k =	2.4536E-01	-8.8366E+00	-9.4452E+00	-9.9221E+00	-1.0000E+00	-2.1644E+00
A4 =	-2.0120E-02	1.4075E-02	9.4789E-02	2.0937E-02	-1.2379E-01	-8.2997E-02
A6 =	2.0705E-01	5.8864E-03	-2.0077E-01	-3.4500E-02	3.4462E-02	2.4327E-02
A8 =	-3.2830E-01	-4.6920E-02	1.7709E-01	6.9060E-03	-3.3927E-03	-5.9157E-03
A10 =	2.5237E-01	3.1447E-02	-1.1395E-01	7.7911E-04	-1.9254E-04	9.8912E-04
A12 =	-1.0838E-01	-5.6323E-03	4.6183E-02	-6.2942E-04	7.4794E-05	-9.9061E-05
A14 =	2.5571E-02	-5.7787E-04	-1.0445E-02	1.0654E-04	-6.3572E-06	5.2884E-06
A16 =	-2.6726E-03	1.8519E-04	9.8878E-04	-6.1029E-06	1.8435E-07	-1.1631E-07

In the 6th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 6th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 11 and Table 12 as the following values and satisfy the following conditions:

6th Embodiment			
f [mm]	5.30	$f/(CT4 + T45 + CT5)$	7.95
Fno	2.32	$R5/ R6 $	0.00
HFOV [deg.]	36.0	$(R11 + R12)/(R11 - R12)$	2.78
$(V2 + V4)/V1$	0.85	$ f3/f4 $	0.29
T34/CT4	2.23	$ f/f3 + f/f4 + f/f5 $	0.70
CT6/CT5	3.05	$f6/f5$	-0.47

7th Embodiment

FIG. 13 is a schematic view of an image capturing device according to the 7th embodiment of the present disclosure. FIG. 14 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 7th embodiment. In FIG. 13, the image capturing device includes the image capturing lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor 790. The image capturing lens assembly includes, in order from an object side to an image side, an aperture stop 700, a first lens element 710, a second lens element 720, a third lens element 730, a fourth lens element 740, a fifth lens element 750, a sixth lens element 760, an IR-cut filter 770 and an image plane 780, wherein the image capturing lens assembly has a total of six lens elements (710-760) with refractive power.

The first lens element 710 with positive refractive power has an object-side surface 711 being convex in a paraxial region thereof and an image-side surface 712 being concave

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in a paraxial region thereof. The first lens element **710** is made of plastic material and has the object-side surface **711** and the image-side surface **712** being both aspheric.

The second lens element **720** with negative refractive power has an object-side surface **721** being convex in a paraxial region thereof and an image-side surface **722** being concave in a paraxial region thereof. The second lens element **720** is made of plastic material and has the object-side surface **721** and the image-side surface **722** being both aspheric.

The third lens element **730** with positive refractive power has an object-side surface **731** being convex in a paraxial region thereof and an image-side surface **732** being convex in a paraxial region thereof. The third lens element **730** is made of plastic material and has the object-side surface **731** and the image-side surface **732** being both aspheric.

The fourth lens element **740** with positive refractive power has an object-side surface **741** being concave in a paraxial region thereof and an image-side surface **742** being convex in a paraxial region thereof. The fourth lens element **740** is made of plastic material and has the object-side surface **741** and the image-side surface **742** being both aspheric.

The fifth lens element **750** with negative refractive power has an object-side surface **751** being convex in a paraxial

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region thereof and an image-side surface **752** being concave in a paraxial region thereof, wherein the image-side surface **752** of the fifth lens element **750** has at least one convex shape in an off-axis region thereof. The fifth lens element **750** is made of plastic material and has the object-side surface **751** and the image-side surface **752** being both aspheric.

The sixth lens element **760** with negative refractive power has an object-side surface **761** being convex in a paraxial region thereof and an image-side surface **762** being concave in a paraxial region thereof, wherein the image-side surface **762** of the sixth lens element **760** has at least one convex shape in an off-axis region thereof. The sixth lens element **760** is made of plastic material and has the object-side surface **761** and the image-side surface **762** being both aspheric.

The IR-cut filter **770** is made of glass and located between the sixth lens element **760** and the image plane **780**, and will not affect the focal length of the image capturing lens assembly. The image sensor **790** is disposed on the image plane **780** of the image capturing lens assembly.

The detailed optical data of the 7th embodiment are shown in Table 13 and the aspheric surface data are shown in Table 14 below.

TABLE 13

7th Embodiment								
f = 5.18 mm, Fno = 2.42, HFOV = 37.0 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.332					
2	Lens 1	1.869 ASP	0.543	Plastic	1.544	55.9	4.01	
3		11.725 ASP	0.163					
4	Lens 2	6.394 ASP	0.300	Plastic	1.639	23.5	-6.20	
5		2.402 ASP	0.243					
6	Lens 3	6.286 ASP	1.190	Plastic	1.544	55.9	8.07	
7		-13.576 ASP	0.372					
8	Lens 4	-2.764 ASP	0.326	Plastic	1.639	23.5	172.87	
9		-2.820 ASP	0.037					
10	Lens 5	2.399 ASP	0.333	Plastic	1.544	55.9	-69.00	
11		2.145 ASP	0.669					
12	Lens 6	7.702 ASP	0.744	Plastic	1.530	55.8	-7.47	
13		2.528 ASP	0.400					
14	IR-cut filter	Plano	0.300	Glass	1.517	64.2	—	
15		Plano	0.391					
16	Image	Plano	—					

Note:

Reference wavelength is 587.6 nm (d-line).

The effective radius of Surface 8 is 1.480 mm.

TABLE 14

Aspheric Coefficients						
Surface #	2	3	4	5	6	7
k =	1.7814E-01	-1.0000E+00	-6.0307E+01	-1.7938E+01	-7.2325E+01	-1.0000E+00
A4 =	-7.5874E-03	-4.4785E-02	-1.1688E-01	1.2130E-02	-1.3549E-02	-2.9098E-02
A6 =	3.4165E-02	9.6347E-02	1.8407E-01	1.9892E-02	2.3970E-02	5.0959E-02
A8 =	-6.4169E-02	-1.2719E-01	-1.8694E-01	8.5426E-04	-9.0145E-02	-9.5656E-02
A10 =	6.6278E-02	1.1542E-01	1.3302E-01	-7.0745E-03	1.6268E-01	9.4782E-02
A12 =	-3.0655E-02	-6.0813E-02	-6.4955E-02	1.3495E-03	-1.6548E-01	-5.3333E-02
A14 =	4.1902E-03	1.1196E-02	1.4315E-02	3.9423E-03	8.9246E-02	1.6231E-02
A16 =	-1.9543E-12	4.8779E-13	-1.6133E-13	3.5336E-14	-1.8600E-02	-1.9306E-03
Surface #	8	9	10	11	12	13
k =	-3.8842E-02	-8.4790E+00	-1.0000E+01	-9.9854E+00	-1.2882E+00	-2.0500E+00
A4 =	5.1554E-03	9.7773E-03	7.7844E-02	5.1195E-02	-1.2520E-01	-1.0474E-01
A6 =	1.1284E-01	-2.2563E-02	-1.6414E-01	-6.9843E-02	2.7040E-02	3.2549E-02
A8 =	-2.3066E-01	-1.3344E-02	1.2811E-01	2.8163E-02	2.0719E-04	-8.3045E-03

TABLE 14-continued

Aspheric Coefficients						
A10 =	2.0018E-01	1.6404E-02	-7.7270E-02	-6.5026E-03	-9.0988E-04	1.4127E-03
A12 =	-8.9692E-02	-2.0323E-03	3.0614E-02	8.5803E-04	1.5149E-04	-1.4371E-04
A14 =	2.1119E-02	-1.0314E-03	-6.8767E-03	-5.8931E-05	-1.0741E-05	7.9781E-06
A16 =	-2.1274E-03	2.0691E-04	6.5014E-04	1.5213E-06	2.9084E-07	-1.8705E-07

In the 7th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 7th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 13 and Table 14 as the following values and satisfy the following conditions:

7th Embodiment			
f [mm]	5.18	$f/(CT4 + T45 + CT5)$	7.44
Fno	2.42	$R5/ R6 $	0.46
HFOV [deg.]	37.0	$(R11 + R12)/(R11 - R12)$	1.98
$(V2 + V4)/V1$	0.84	$ f3/f4 $	0.05
T34/CT4	1.14	$ f/f3 + f/f4 + f/f5 $	0.75
CT6/CT5	2.23	$f6/f5$	-0.11

8th Embodiment

FIG. 15 is a schematic view of an image capturing device according to the 8th embodiment of the present disclosure. FIG. 16 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 8th embodiment. In FIG. 15, the image capturing device includes the image capturing lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor 890. The image capturing lens assembly includes, in order from an object side to an image side, an aperture stop 800, a first lens element 810, a second lens element 820, a third lens element 830, a fourth lens element 840, a fifth lens element 850, a sixth lens element 860, an IR-cut filter 870 and an image plane 880, wherein the image capturing lens assembly has a total of six lens elements (810-860) with refractive power.

The first lens element 810 with positive refractive power has an object-side surface 811 being convex in a paraxial region thereof and an image-side surface 812 being concave in a paraxial region thereof. The first lens element 810 is made of plastic material and has the object-side surface 811 and the image-side surface 812 being both aspheric.

The second lens element 820 with negative refractive power has an object-side surface 821 being convex in a paraxial region thereof and an image-side surface 822 being concave in a paraxial region thereof. The second lens element 820 is made of plastic material and has the object-side surface 821 and the image-side surface 822 being both aspheric.

The third lens element 830 with positive refractive power has an object-side surface 831 being convex in a paraxial region thereof and an image-side surface 832 being convex in a paraxial region thereof. The third lens element 830 is made of plastic material and has the object-side surface 831 and the image-side surface 832 being both aspheric.

The fourth lens element 840 with positive refractive power has an object-side surface 841 being concave in a paraxial region thereof and an image-side surface 842 being convex in a paraxial region thereof. The fourth lens element 840 is made of plastic material and has the object-side surface 841 and the image-side surface 842 being both aspheric.

The fifth lens element 850 with positive refractive power has an object-side surface 851 being convex in a paraxial region thereof and an image-side surface 852 being concave in a paraxial region thereof, wherein the image-side surface 852 of the fifth lens element 850 has at least one convex shape in an off-axis region thereof. The fifth lens element 850 is made of plastic material and has the object-side surface 851 and the image-side surface 852 being both aspheric.

The sixth lens element 860 with negative refractive power has an object-side surface 861 being concave in a paraxial region thereof and an image-side surface 862 being concave in a paraxial region thereof, wherein the image-side surface 862 of the sixth lens element 860 has at least one convex shape in an off-axis region thereof. The sixth lens element 860 is made of plastic material and has the object-side surface 861 and the image-side surface 862 being both aspheric.

The IR-cut filter 870 is made of glass and located between the sixth lens element 860 and the image plane 880, and will not affect the focal length of the image capturing lens assembly. The image sensor 890 is disposed on the image plane 880 of the image capturing lens assembly.

The detailed optical data of the 8th embodiment are shown in Table 15, and the aspheric surface data are shown in Table 16 below.

TABLE 15

8th Embodiment								
f = 5.22 mm, Fno = 2.42, HFOV = 37.0 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.347					
2	Lens 1	1.893	ASP	0.523	Plastic	1.544	55.9	4.09
3		11.367	ASP	0.094				
4	Lens 2	5.706	ASP	0.258	Plastic	1.639	23.5	-6.73
5		2.410	ASP	0.352				
6	Lens 3	6.109	ASP	1.083	Plastic	1.544	55.9	8.20

TABLE 15-continued

8th Embodiment f = 5.22 mm, Fno = 2.42, HFOV = 37.0 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
7		-15.534	ASP	0.326				
8	Lens 4	-2.345	ASP	0.346	Plastic	1.639	23.5	172.55
9		-2.429	ASP	0.048				
10	Lens 5	1.736	ASP	0.245	Plastic	1.544	55.9	321.69
11		1.666	ASP	0.855				
12	Lens 6	-64.206	ASP	0.604	Plastic	1.530	55.8	-7.93
13		4.512	ASP	0.500				
14	IR-cut filter	Plano	0.175	Glass	1.517	64.2		—
15		Plano	0.538					
16	Image	Plano	—					

Note:

Reference wavelength is 587.6 nm (d-line).

The effective radius of Surface 10 is 1.710 mm.

TABLE 16

Aspheric Coefficients						
Surface #	2	3	4	5	6	7
k =	2.0396E-01	1.4861E+00	-5.3648E+01	-1.7706E+01	-7.2362E+01	-1.0000E+00
A4 =	-5.5636E-03	-4.2025E-02	-1.1452E-01	8.8324E-03	2.1715E-03	-7.1293E-02
A6 =	3.1996E-02	1.0086E-01	1.8717E-01	1.7191E-02	-9.2222E-03	2.9169E-01
A8 =	-6.3871E-02	-1.2392E-01	-1.8445E-01	-1.1925E-03	-2.6075E-02	-7.1304E-01
A10 =	6.7418E-02	1.1612E-01	1.3378E-01	-7.0897E-03	7.4185E-02	9.1511E-01
A12 =	-3.0239E-02	-6.1810E-02	-6.5435E-02	4.1982E-03	-9.8226E-02	-6.4273E-01
A14 =	4.1734E-03	1.1196E-02	1.4315E-02	3.9423E-03	6.6317E-02	2.3320E-01
A16 =	-2.8070E-09	-1.6796E-09	2.5057E-09	-5.9162E-10	-1.6057E-02	-3.3731E-02
Surface #	8	9	10	11	12	13
k =	-1.0406E+00	-8.4183E+00	-8.3724E+00	-9.9856E+00	-1.0000E+00	-2.0666E+00
A4 =	-1.1415E-02	1.9470E-02	1.7973E-01	1.8683E-01	-1.0094E-01	-8.9496E-02
A6 =	3.0723E-01	-1.7552E-02	-3.0857E-01	-2.7705E-01	9.7914E-03	2.3706E-02
A8 =	-7.2711E-01	-8.1351E-02	1.7886E-01	1.5164E-01	-9.8912E-03	-1.1404E-02
A10 =	7.9675E-01	9.9891E-02	-7.1716E-02	-4.6889E-02	8.2067E-03	3.8507E-03
A12 =	-4.6445E-01	-4.1158E-02	2.2771E-02	8.3952E-03	-2.3236E-03	-6.8410E-04
A14 =	1.4233E-01	7.1339E-03	-5.1146E-03	-8.1051E-04	2.8897E-04	5.9148E-05
A16 =	-1.8187E-02	-4.4460E-04	5.2913E-04	3.1870E-05	-1.3645E-05	-1.9572E-06

In the 8th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 8th embodiment, so an explanation in this regard will not be provided again. Moreover, these parameters can be calculated from Table 15 and Table 16 as the following values and satisfy the following conditions:

8th Embodiment			
f [mm]	5.22	f/(CT4 + T45 + CT5)	8.17
Fno	2.42	R5/ R6	0.39
HFOV [deg.]	37.0	(R11 + R12)/(R11 - R12)	0.87
(V2 + V4)/V1	0.84	f3/f4	0.05
T34/CT4	0.94	f/f3 + f/f4 + f/f5	0.68
CT6/CT5	2.47	f6/f5	-0.02

The foregoing image capturing device **10** may be installed in but not limited to a mobile terminal, including a smart phone (FIG. 17), a tablet personal computer (FIG. 18) or a wearable device (FIG. 19). Moreover, the image capturing device **10** includes an image capturing lens assembly (not otherwise herein labeled) and an image sensor (not otherwise

herein labeled), wherein the image sensor is disposed on an image plane of the image capturing lens assembly.

The foregoing description, for purpose of explanation, has been described with reference to specific embodiments. It is to be noted that TABLES 1-16 show different data of the different embodiments; however, the data of the different embodiments are obtained from experiments. The embodiments were chosen and described in order to best explain the principles of the disclosure and its practical applications, to thereby enable others skilled in the art to best utilize the disclosure and various embodiments with various modifications as are suited to the particular use contemplated. The embodiments depicted above and the appended drawings are exemplary and are not intended to be exhaustive or to limit the scope of the present disclosure to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings.

What is claimed is:

1. An image capturing lens assembly comprising, in order from an object side to an image side:

a first lens element with positive refractive power having an object-side surface being convex in a paraxial region thereof;

a second lens element having refractive power;

a third lens element having refractive power;

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a fourth lens element with refractive power having an object-side surface being concave in a paraxial region thereof;

a fifth lens element with refractive power having an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof, wherein the image-side surface of the fifth lens element has at least one convex shape in an off-axis region thereof, and the object-side surface and the image-side surface of the fifth lens element are aspheric; and

a sixth lens element with refractive power having an image-side surface being concave in a paraxial region thereof, wherein the image-side surface of the sixth lens element has at least one convex shape in an off-axis region thereof, and an object-side surface and the image-side surface of the sixth lens element are aspheric;

wherein the image capturing lens assembly has a total of six lens elements with refractive power, a focal length of the third lens element is f_3 , a focal length of the fourth lens element is f_4 , an axial distance between the third lens element and the fourth lens element is T_{34} , a central thickness of the fourth lens element is CT_4 , a central thickness of the fifth lens element is CT_5 , a central thickness of the sixth lens element is CT_6 , and the following conditions are satisfied:

$$|f_3/f_4| < 0.50;$$

$$0.9 < T_{34}/CT_4; \text{ and}$$

$$1.6 < CT_6/CT_5.$$

2. The image capturing lens assembly of claim 1, wherein a curvature radius of an object-side surface of the third lens element is R_5 , a curvature radius of an image-side surface of the third lens element is R_6 , and the following condition is satisfied:

$$R_5/|R_6| < 1.0.$$

3. The image capturing lens assembly of claim 1, wherein a focal length of the image capturing lens assembly is f , the focal length of the third lens element is f_3 , the focal length of the fourth lens element is f_4 , a focal length of the fifth lens element is f_5 , and the following condition is satisfied:

$$0.20 < |f/f_3| + |f/f_4| + |f/f_5| < 1.0.$$

4. The image capturing lens assembly of claim 1, wherein the fifth lens element has negative refractive power.

5. The image capturing lens assembly of claim 1, wherein a focal length of the fifth lens element is f_5 , a focal length of the sixth lens element is f_6 , and the following condition is satisfied:

$$-0.65 < f_6/f_5 < 0.$$

6. The image capturing lens assembly of claim 1, wherein a focal length of the image capturing lens assembly is f , the central thickness of the fourth lens element is CT_4 , the central thickness of the fifth lens element is CT_5 , an axial distance between the fourth lens element and the fifth lens element is T_{45} , and the following condition is satisfied:

$$5 \leq f/(CT_4 + T_{45} + CT_5) < 10.$$

7. An image capturing device, comprising:
the image capturing lens assembly of claim 1; and
an image sensor, wherein the image sensor is located on an image plane of the image capturing lens assembly.

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8. A mobile terminal, comprising:

the image capturing device of claim 7.

9. The image capturing lens assembly of claim 1, wherein the second lens element has negative refractive power.

10. The image capturing lens assembly of claim 9, wherein a curvature radius of the object-side surface of the sixth lens element is R_{11} , a curvature radius of the image-side surface of the sixth lens element is R_{12} , and the following condition is satisfied:

$$0 < (R_{11} + R_{12}) / (R_{11} - R_{12}).$$

11. The image capturing lens assembly of claim 9, wherein the object-side surface of the sixth lens element is convex in a paraxial region thereof.

12. The image capturing lens assembly of claim 11, wherein a focal length of the image capturing lens assembly is f , the focal length of the third lens element is f_3 , the focal length of the fourth lens element is f_4 , a focal length of the fifth lens element is f_5 , and the following condition is satisfied:

$$0.20 < |f/f_3| + |f/f_4| + |f/f_5| < 1.35.$$

13. The image capturing lens assembly of claim 11, wherein an Abbe number of the first lens element is V_1 , an Abbe number of the second lens element is V_2 , an Abbe number of the fourth lens element is V_4 , and the following condition is satisfied:

$$0.6 < (V_2 + V_4) / V_1 < 1.2.$$

14. The image capturing lens assembly of claim 11, wherein the axial distance between the third lens element and the fourth lens element is T_{34} , the central thickness of the fourth lens element is CT_4 , and the following condition is satisfied:

$$1.25 < T_{34}/CT_4 < 2.75.$$

15. The image capturing lens assembly of claim 11, wherein the central thickness of the fifth lens element is CT_5 , the central thickness of the sixth lens element is CT_6 , and the following condition is satisfied:

$$2.0 < CT_6/CT_5 < 4.0.$$

16. An image capturing lens assembly comprising, in order from an object side to an image side:

a first lens element with positive refractive power having an object-side surface being convex in a paraxial region thereof;

a second lens element having refractive power;

a third lens element having refractive power;

a fourth lens element with refractive power having an object-side surface being concave in a paraxial region thereof;

a fifth lens element with refractive power having an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof, wherein the image-side surface of the fifth lens element has at least one convex shape in an off-axis region thereof, and the object-side surface and the image-side surface of the fifth lens element are aspheric; and

a sixth lens element with negative refractive power having an image-side surface being concave in a paraxial region thereof, wherein the image-side surface of the sixth lens element has at least one convex shape in an off-axis region thereof, and an object-side surface and the image-side surface of the sixth lens element are aspheric;

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wherein the image capturing lens assembly has a total of six lens elements with refractive power, a focal length of the third lens element is f_3 , a focal length of the fourth lens element is f_4 , a focal length of the fifth lens element is f_5 , a focal length of the sixth lens element is f_6 , an axial distance between the third lens element and the fourth lens element is T_{34} , a central thickness of the fourth lens element is CT_4 , and the following conditions are satisfied:

$$|f_3/f_4| < 0.50;$$

$$0.9 < T_{34}/CT_4; \text{ and}$$

$$-0.65 < f_6/|f_5| < 0.$$

17. The image capturing lens assembly of claim 16, wherein the third lens element has positive refractive power, a curvature radius of an object-side surface of the third lens element is R_5 , a curvature radius of an image-side surface of the third lens element is R_6 , and the following condition is satisfied:

$$R_5/|R_6| < 1.0.$$

18. The image capturing lens assembly of claim 16, wherein the focal length of the fifth lens element is f_5 , the focal length of the sixth lens element is f_6 , and the following condition is satisfied:

$$-0.50 < f_6/|f_5| < 0.$$

19. The image capturing lens assembly of claim 16, wherein a focal length of the image capturing lens assembly is f , the focal length of the third lens element is f_3 , the focal

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length of the fourth lens element is f_4 , the focal length of the fifth lens element is f_5 , and the following condition is satisfied:

$$0.20 < |f/f_3| + |f/f_4| + |f/f_5| < 1.35.$$

20. The image capturing lens assembly of claim 16, wherein an Abbe number of the first lens element is V_1 , an Abbe number of the second lens element is V_2 , and an Abbe number of the fourth lens element is V_4 , the following condition is satisfied:

$$0.6 < (V_2 + V_4)/V_1 < 1.2.$$

21. An image capturing device, comprising:
the image capturing lens assembly of claim 16; and
an image sensor, wherein the image sensor is located on an image plane of the image capturing lens assembly.

22. A mobile terminal, comprising:
the image capturing device of claim 21.

23. The image capturing lens assembly of claim 16, wherein the second lens element has negative refractive power, an object-side surface of the second lens element being convex in a paraxial region thereof, and an image-side surface of the second lens element being concave in a paraxial region thereof.

24. The image capturing lens assembly of claim 23, wherein the axial distance between the third lens element and the fourth lens element is T_{34} , the fourth lens element is CT_4 , and the following condition is satisfied:

$$1.25 < T_{34}/CT_4 < 2.75.$$

25. The image capturing lens assembly of claim 23, wherein a focal length of the image capturing lens assembly is f , the central thickness of the fourth lens element is CT_4 , a central thickness of the fifth lens element is CT_5 , an axial distance between the fourth lens element and the fifth lens element is T_{45} , and the following condition is satisfied:

$$5 \leq f/(CT_4 + T_{45} + CT_5) < 10.$$

* * * * *